
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) has become one of the most widely adopted neighbor-embedding methods, deriving its low-dimensional representations from the interplay of attractive and repulsive forces among high-dimensional data points. We decompose these forces into their scalar, distance-dependent functional constituents—which we term the attraction and repulsion *shapes*—and derive the exact per-update conditions under which each contracts or expands the distance separating a pair of points. This decomposition furnishes a single analytical lens through which the behavior of UMAP, and of the wider family of neighbor-embedding methods, becomes legible: repulsion accentuates dissimilarity and thereby regulates cluster boundaries and inter-cluster separation, whereas attraction operates far more subtly, in that the attractive update between a pair of points can effect either contraction or expansion depending on their separation in the low-dimensional map. This duality accounts for the necessity of learning-rate annealing and explains why the attractive and repulsive terms warrant distinct treatment. Building directly on the analysis, we modify the attraction shape and, across three datasets, markedly improve the consistency of the resulting embeddings under random initialization, as quantified by Procrustes distance aggregated over many runs. Collectively, the shape formalism renders UMAP and its relatives more interpretable and their embeddings measurably more consistent and more controllable.

1 Introduction

The contemporary scientific landscape is defined by an ever-growing abundance of high-dimensional data, and dimensionality reduction (DR) has become an indispensable instrument for exploratory analysis, rendering the latent structure of such data visible through low-dimensional visualization. The two dominant methods in this space, t -distributed stochastic neighbor embedding (t -SNE) (Maaten and Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018), are both grounded in attraction–repulsion dynamics that draw similar data points together while driving dissimilar ones apart. As unsupervised algorithms, they dispense with labeled data entirely, recovering and preserving the intrinsic structure of high-dimensional data by exploiting local (attractive) and global (repulsive) relationships among points. This capacity for label-free structure recovery makes them well suited to a broad spectrum of tasks in which the visualization of latent patterns in unlabeled data is both critical and informative, spanning clustering (Becht et al., 2019), exploratory data analysis (Fleischer and Islam, 2020), anomaly detection in semiconductor manufacturing (Fan et al., 2021), visual search (González-Márquez et al., 2024), time-series analysis (Altin and Cakir, 2024), the study of representation convergence (Huh et al., 2024), and outlier image detection (Islam and Fleischer, 2024). In distilling complex data into readily interpretable form through a purely data-driven optimization, DR thus exemplifies the broader capacity of unsupervised learning.

Building upon this attraction–repulsion principle, a succession of newer methods has emerged (Amid and Warmuth, 2019; Agrawal et al., 2021; Wang et al., 2021; Narayan et al., 2021; Yang et al., 2022; Wang et al., 2025; Kury et al., 2025), each engineered to emphasize a particular aspect of the data. For all their breadth of application, these methods rest heavily on heuristic practice that can yield embeddings resistant to meaningful interpretation, and the distortions that DR introduces are provably unavoidable (Chari and Pachter, 2023). A mechanistic understanding of these algorithms is therefore not a luxury but a prerequisite: it is what allows practitioners to interpret embeddings soundly, to guard against spurious structure, and to optimize performance deliberately rather than by trial and error. In practice, compact clusters are coaxed out through an assortment of techniques—bespoke initialization, learning-rate scheduling, and kernel-function tuning—yet the attractive and repulsive dynamics that actually drive cluster formation have never

been examined in depth. Compounding the difficulty, the parameters that govern these dynamics remain buried within abstract functional forms, obscuring any principled account of how the algorithms behave.

In this paper, we decompose these forces into their constituent parts and extract the functional *shapes* of attraction and repulsion—scalar coefficients that depend only on the low-dimensional pairwise distance. Analyzing UMAP through these shapes traces three phenomena long treated as separate—the necessity of learning-rate annealing, the difficulty of producing consistent output under random initialization, and the mechanism of cluster formation—to a common source in the attraction shape, while identifying inter-cluster distance as the province chiefly of repulsion. Viewing the forces through their shapes thus supplies a single, unified account of behaviors that the literature has previously explained only in isolation. Our specific contributions are as follows:

1. We formalize the attractive and repulsive forces as distance-dependent *shapes* and derive the exact per-update conditions under which each contracts or expands a pairwise distance, furnishing a unifying analytical perspective that applies, without modification, to any method governed by attraction and repulsion (Section 4).
2. We show that UMAP’s default attraction shape produces the counterintuitive coexistence of contraction and expansion within a single attractive update, and, by comparing the attraction shapes of competing algorithms, we identify this feature as the precise reason UMAP—unlike NEG-*t*-SNE or PaCMAP—requires learning-rate annealing (Section 5).
3. We modify the attraction shape to strengthen attraction at large distances and thereby substantially improve the consistency of embeddings under random initialization, cutting the mean Procrustes distance to a PCA-initialized reference by roughly a third to a half across three datasets and providing direct evidence that UMAP encodes an underlying structure largely independent of its initialization (Section 5.1).
4. Through the repulsion shape, we relate cluster compactness and inter-cluster distance to repulsion, and we introduce a direct means of controlling that distance—an additive offset to the repulsion shape—that reaches regimes inaccessible to conventional parameter tuning (Section 5.2).
5. We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG-*t*-SNE, make explicit the similarities and distinctions among them—including a formal equivalence between Parametric UMAP and NEG-*t*-SNE—and relate the learning-rate sensitivity of each method to the range of its attraction shape (Section 6).

We center the main text on UMAP, now the de facto standard across many fields, and introduce the other algorithms wherever they illuminate the analysis; the appendices develop each in detail, extending the same formalism to Parametric UMAP, PaCMAP, LocalMAP, TriMap, *t*-SNE, SNE, and MDS. The code used in this research will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor-embedding algorithms originate with stochastic neighbor embedding (SNE) (Hinton and Roweis, 2002) and its heavy-tailed extension, *t*-SNE (Maaten and Hinton, 2008). Both construct a dense graph in which every point maintains a pairwise relation to every other, similar or not, and both normalize the graph weights into a probability distribution. Contemporaneous methods—locally linear embedding (Roweis and Saul, 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi, 2002)—instead operate on a *k*-nearest-neighbor (*k*-NN) graph of pairwise interactions and, as spectral methods, obtain the embedding through eigenvalue decomposition. The pivotal transition toward the modern formulation came when Tang et al. (2016) embedded the *k*-NN graph within an iterative optimization and discarded normalization in the low dimension. McInnes et al. (2018) carried this trajectory to its conclusion in UMAP, dispensing with normalization entirely—in both the high and the low dimension—and recovering the embedding from pairwise interactions alone, through an explicit attractive force that preserves local neighborhoods and a repulsive force that holds dissimilar points apart. Subsequent methods such as PaCMAP (Wang et al., 2021) and NEG-*t*-SNE (Damrich et al., 2023) build directly on these foundations; for a recent survey, see de Bodt et al. (2025).

Considerable progress has been made in understanding and formally relating these algorithms. An early analysis of SNE established that data well-clustered in the original space remain well-clustered in the embedding (Shaham and Steinerberger, 2017), and the parallel analysis of *t*-SNE by Linderman and Steinerberger (2019) proved that the number of clusters in the embedding lower-bounds the number in the original space, a line subsequently sharpened by further characterizations (Arora et al., 2018; Cai and Ma, 2022; Linderman and Steinerberger, 2022). Because *t*-SNE and UMAP arise from a common framework yet yield markedly different visualizations, reconciling the two has been a central undertaking in the literature (Böhm et al., 2022; Damrich et al., 2023; Draganov et al., 2023). Böhm et al. (2022) positioned Laplacian Eigenmaps, ForceAtlas2 (Jacomy et al., 2014), UMAP, and *t*-SNE as samples from a single

underlying spectrum; LE and t -SNE are linked through the early-exaggeration phase (Cai and Ma 2022), and UMAP and t -SNE through contrastive estimation (Damrich et al. 2023). Independently and contemporaneously, Hu et al. (2023) uncovered the relationship between contrastive learning and SNE. More recent work explains dimensionality reduction from a probabilistic standpoint (Ravuri et al. 2023; Ravuri and Lawrence 2024), through kernel techniques (Draganov and Dohn 2023), and via information geometry (Kolpakov and Rocke 2024). Whereas these accounts relate whole loss functions or entire embeddings, our analysis operates at the finer granularity of the attractive and repulsive forces themselves, and it is at this granularity that the mechanisms distinguishing the methods become apparent.

3 Uniform Manifold Approximation and Projection

UMAP constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ through a pairwise affinity $p_{i,j} = f_h(d(x_i, x_j))$ normalized to lie in $[0, 1]$; the full construction is deferred to Appendix H

The corresponding low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is equipped with a differentiable affinity function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

in which the parameters a and b set the density of the mapping and are obtained by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d, \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where the minimum-distance parameter m_d regulates the separation between the two nearest low-dimensional points. UMAP then obtains the embedding by minimizing the cross-entropy loss between the high- and low-dimensional affinities,

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

Here the first term supplies an attractive force and the second a repulsive force. Rather than update every point at each iteration, UMAP adopts the negative-sampling strategy (Mikolov et al. 2013; Tang et al. 2016): for each edge with $p_{ij} > 0$ —a *positive* edge—a set of edges is drawn at random to serve as *negative* edges, so that the two terms of the loss are optimized on disjoint edge sets. The attractive force acts on the positive edge,

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

while the repulsive force acts on the negative edges,

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t indexes the optimization step. Crucially, for a negative edge y_j is held fixed while y_i alone is updated—an asymmetry that our subsequent analysis makes essential. A detailed analysis of UMAP’s loss is given by Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

The action of updates (4, 6) admits a considerable simplification once each gradient— $\nabla_{y_i^t} \log(q_{ij})$ for attraction and $\nabla_{y_i^t} \log(1 - q_{ij})$ for repulsion—is decomposed into a scalar coefficient, dependent solely on the distance $\|y_i - y_j\|_2$, acting on the displacement vector $(y_i - y_j)$. We term this scalar coefficient the attraction (respectively, repulsion) *shape*. The decomposition is exact and, though we develop it through UMAP, it applies without modification to any method whose updates are governed by attractive and repulsive forces—a generality we exploit throughout the appendices.

Writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$ with $\zeta = \|y_i - y_j\|_2$, the update equations for a positive edge take the form

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and we have used the fact that, under the Euclidean metric, $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$, so that a positive edge displaces its two endpoints symmetrically. The update for a negative edge admits the analogous form

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Decompositions of this kind have appeared before (McInnes et al., 2018; Agrawal et al., 2021; Draganov and Dohn, 2023), but their formulation and purpose differ fundamentally from ours. The original UMAP paper (McInnes et al., 2018) employed the decomposition as a computational device for fast processing, and Draganov and Dohn (2023) used it to compare algorithms from a kernel perspective; in both cases the coefficient was a means to computing the derivative. We instead treat the two coefficients as functions in their own right—objects that may assume a variety of forms and be reasoned about, and redesigned, directly. Agrawal et al. (2021) took a related step earlier, but expressed the decomposition as a scalar coefficient multiplying the *unit* vector $(y_i - y_j)/\|y_i - y_j\|_2$, thereby isolating force magnitude alone. Retaining the full displacement vector rather than its normalization is what makes our formulation more informative: as the following analysis shows, it preserves the sign structure that a magnitude-only view discards, and it is precisely this sign structure that dictates whether an update contracts or expands.

4.1 Conditions for Attraction and Repulsion

We now derive, directly from the update equations (7)–(10), the exact conditions under which a single update contracts or expands a pairwise distance. The first proposition characterizes contraction of the distance between the endpoints y_i^t and y_j^t of a positive edge.

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

The product λf_a thus acts as the *effective* attraction shape—the object whose sign and magnitude govern the outcome of an update—although in most cases f_a alone suffices to draw meaningful conclusions about the resulting embeddings. The companion proposition characterizes expansion of the distance between the endpoints y_i^t and y_j^t of a negative edge.

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Including a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) leaves this conclusion intact. We emphasize the precise scope of Propositions 4.1 and 4.2: each is a per-iteration certificate that a *single* isolated update contracts or expands the distance of one pair. They are not global statements about the embedding, and their value lies in what they license downstream—because contraction and expansion can be guaranteed for a chosen distance regime independently of any learning-rate schedule, the shapes become objects one can deliberately design rather than tune by trial and error. The passage from these local certificates to the aggregate behavior of many simultaneous updates is taken up empirically in the sections that follow.

Figure 1(a) illustrates the effect of varying $f_a < 0$ and $f_r > 0$ on a pair of points. The repulsion shape behaves transparently: any positive value increases the distance. The attraction shape is considerably richer, because f_a encodes both contractive and expansive dynamics within a single sign-definite coefficient. For $f_a \in (-1.0, 0)$ the distance contracts, with maximum attraction—coincidence of the two points—attained at $f_a = -0.5$; at yet more negative values the endpoints overshoot and their order flips, and for any $f_a < -1.0$ the distance instead expands. Each update thus acts strictly locally on a single pair, yet the cumulative effect of many such updates is what shapes the global structure of the embedding—a passage from local rule to global outcome that motivates the empirical analysis below.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Applying the gradient decomposition to the low-dimensional affinity (1) yields the attraction and repulsion shapes in closed form,

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

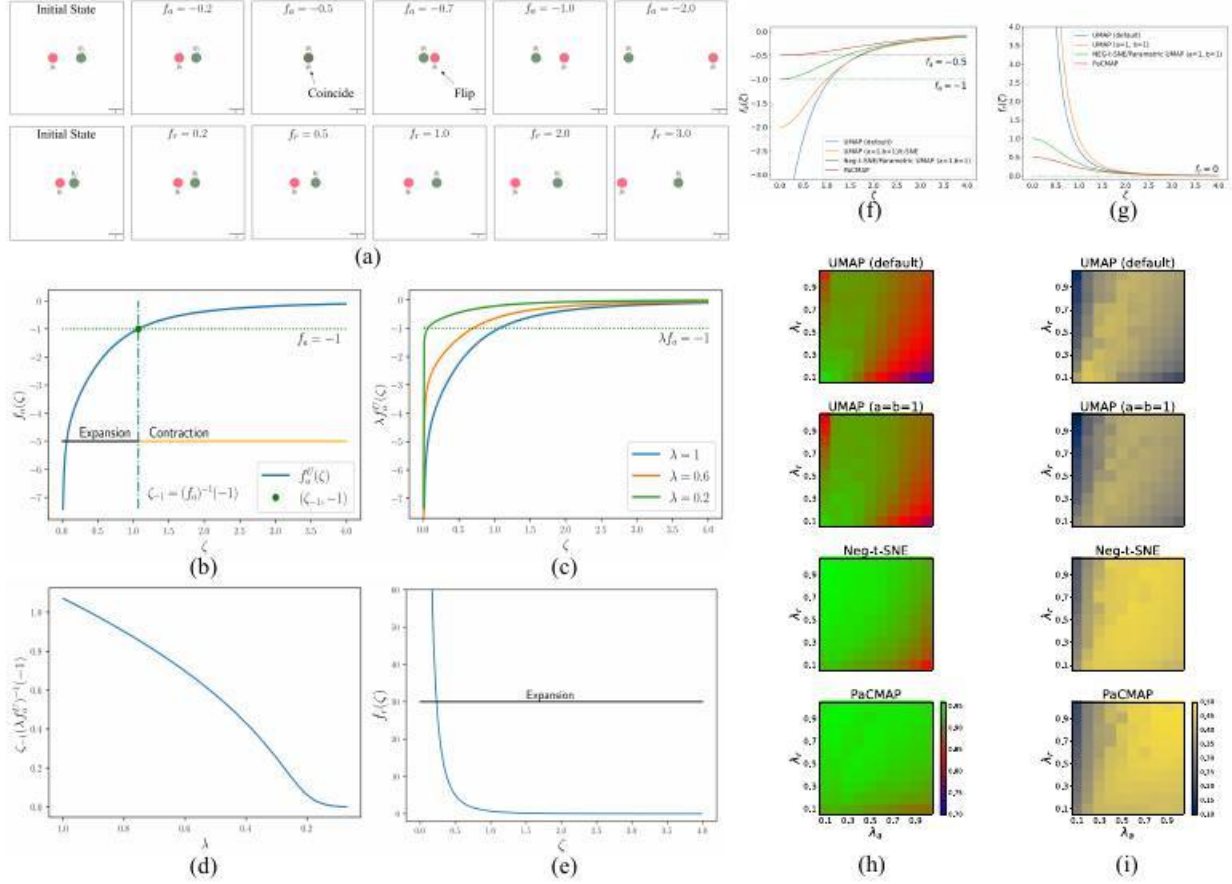


Figure 1: Attraction and repulsion shapes in UMAP. (a) Effect of varying the attraction shape f_a (top) and repulsion shape f_r (bottom) on a single pair of points. (b) Attraction shape of UMAP, which diverges as the separation $\zeta \rightarrow 0$. (c) Effective attraction shape (λf_a^U) across learning rates λ : decreasing λ pulls the shape into the contractive regime $(-1, 0)$. (d) The minimum distance for contraction ζ_{-1} falls toward zero as λ decreases. (e) Repulsion shape of UMAP. (f) Attraction and (g) repulsion shapes of the compared embedding methods. (h) Trustworthiness and (i) silhouette score on MNIST as the constant learning rate is varied independently for attraction (λ_a) and repulsion (λ_r); of the methods compared, only UMAP requires a small λ_a to embed well. Default UMAP parameters throughout: $a = 1.58$, $b = 0.89$.

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section examines the default shapes of UMAP, governed principally by the parameters a and b , together with the insights that emerge from perturbing them. Throughout, we restrict attention to $b \leq 1$, the regime in which the attraction shape is strictly increasing and therefore invertible—a property on which the remaining analysis relies (the derivation, including the boundary at $b = 1$, is given in Appendix A).

Figure 1(b) displays the default attraction shape of UMAP ($a = 1.58$, $b = 0.89$), which diverges to $-\infty$ as $\zeta \rightarrow 0$. By Proposition 4.1 the transition from contraction to expansion occurs precisely where the effective shape λf_a^U crosses -1 , which happens as ζ approaches 0. Because the shape is invertible, this transition is located at a well-defined distance, $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, which admits a clean interpretation as the *minimum distance for contraction* under attractive updates: pairs with separation $\zeta > \zeta_{-1}$ contract, whereas those with $0 < \zeta < \zeta_{-1}$ expand. This overturns the prevailing intuition that an attractive update invariably draws a pair together; for sufficiently close pairs, UMAP’s attractive term does the opposite and pushes them apart.

When ζ_{-1} is large, neighboring pairs oscillate between contraction and expansion and the clusters they form appear fuzzy. The pursuit of sharp cluster boundaries can therefore be recast as a single, concrete objective: driving $\zeta_{-1} \rightarrow 0$. This reframing exposes the role of UMAP’s learning-rate schedule, for annealing λ toward zero is precisely the mechanism that collapses ζ_{-1} (Figs. 1(c,d))—a diagnosis we corroborate empirically below. The repulsion shape

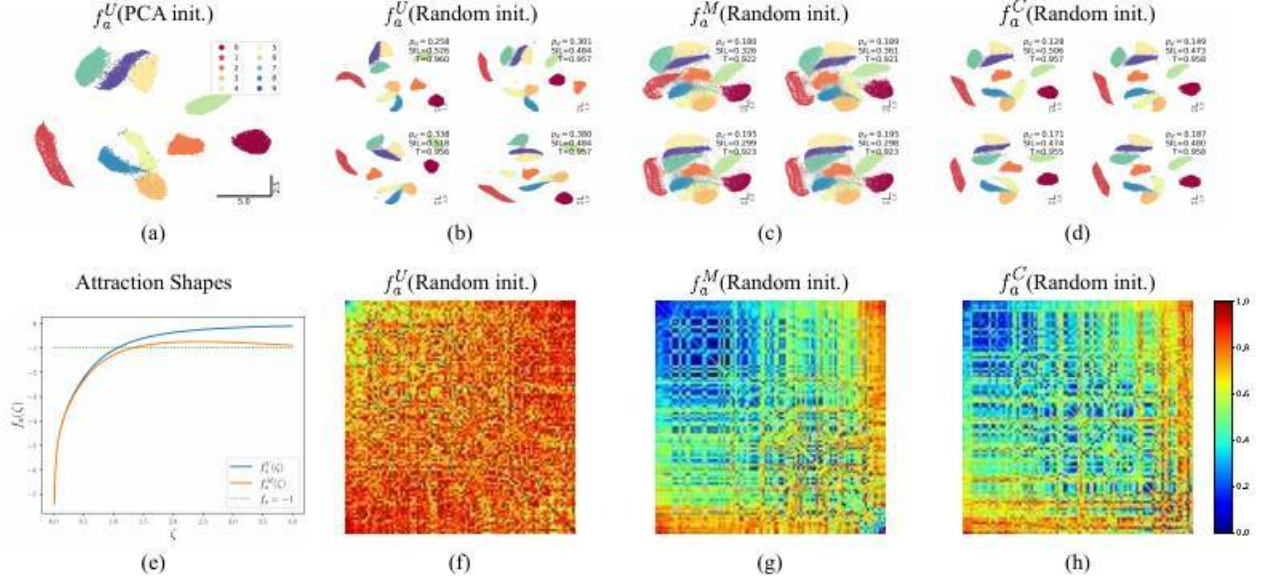


Figure 2: Effect of random initialization on different attraction shapes for the MNIST dataset. (a) Reference mapping obtained with PCA initialization. (b-d) The four mappings with the lowest Procrustes distance (p_d) from the reference (a), for the (b) default UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices aggregated over 100 runs for the (f) default UMAP ($p_d = 0.78 \pm 0.13$), (g) modified (0.49 ± 0.21), and (h) composite (0.50 ± 0.20) attraction shapes (mean $\pm$ std over the run pairs). The diagonal (i, i) entries are sorted by p_d from the reference (a), while each off-diagonal (i, j) entry reports the p_d between the i^{th} and j^{th} random-initialization mappings. Matrices and summary statistics agree: default UMAP embeddings are largely dissimilar across runs, whereas the modified and composite shapes—which reduce the mean p_d from 0.78 to roughly 0.50—recover a mapping that is consistent up to the rotation, reflection, and scaling that Procrustes alignment absorbs.

(Eq. 12), by contrast, is strictly positive and therefore satisfies Proposition 4.2 unconditionally; f_r^U vanishes as $\zeta \rightarrow \infty$ and diverges as $\zeta \rightarrow 0$ (Fig. 1(e)), so that repulsion, unlike attraction, never changes sign.

Comparing attraction shapes across algorithms clarifies why this issue is peculiar to certain methods rather than generic to neighbor embedding. Only UMAP and t -SNE must contend with $f_a < -1$ (and hence $\zeta_{-1} > 0$ at $\lambda = 1$; Fig. 1(f)): t -SNE neutralizes it by weighting each update with the corresponding p_{ij} , whereas UMAP relies on learning-rate annealing. NEG- t -SNE and PaCMAP, by contrast, confine f_a to $[-1, 0]$ by construction, thereby satisfying Proposition 4.1 for every $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$; PaCMAP’s weighted f_a , bounded below 0.5 in magnitude, goes further still and precludes any sign flip during attraction. The repulsion shapes tell a complementary story: of the methods compared, only UMAP attains large values at small distances (Fig. 1(g)), where its shape is formally unbounded, though in practice these values are clipped during optimization. Appendix G treats these and related methods in full detail within the same formalism.

To test this account against measured embedding quality, we assign separate constant learning rates to attraction (λ_a) and repulsion (λ_r) and sweep each independently on MNIST (Figs. 1(h,i)). For UMAP, only a small λ_a (below 0.5, so that $\zeta_{-1} \simeq 0$) yields high-quality embeddings under both metrics: trustworthiness (T) (Venna and Kaski 2001), which quantifies local-neighborhood preservation (Fig. 1(h)), and the silhouette score (SIL) (Rousseeuw 1987), which quantifies label separation (Fig. 1(i)); the repulsion rate, meanwhile, remains admissible across the full range $\lambda_r \in [0, 1]$. For NEG- t -SNE and PaCMAP, both of which enforce $\zeta_{-1} = 0$ by construction, the entire range of λ_a is effective. These sweeps substantiate the mechanism at the level of embedding quality: on this dataset and these metrics, UMAP embeds and clusters well only when ζ_{-1} is forced close to zero—which in practice it achieves through learning-rate annealing—whereas the newer algorithms achieve $\zeta_{-1} = 0$ from the outset and are correspondingly insensitive to the attraction learning rate. Further discussion, results, and illustrations across additional datasets appear in Section 6 and Appendices B, G, and I.

5.1 Improving UMAP’s Consistency under Random Initialization

The consistency of UMAP embeddings depends critically on initialization (Kobak and Linderman, 2021; Wang et al., 2021). Conventionally, the embedding is initialized from a principal component analysis (PCA) of the data or from a spectral decomposition of the high-dimensional graph, either of which yields consistent mappings despite the several sources of stochasticity in the optimization. Under random initialization, by contrast, clusters frequently fail to form at all, or form in an arbitrary orientation that differs from run to run. The shape formalism renders the cause transparent: when two high-dimensional nearest neighbors are initialized far apart, the attractive force between them is too weak to draw them together within the optimization budget. This phenomenon—near-sightedness (Wang et al., 2021)—follows directly from the attraction shape, whose magnitude $|f_a^U|$ decays toward zero with increasing distance rapidly enough that $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)|\zeta = 0$; the attractive displacement it induces therefore vanishes at long range.

This diagnosis points directly to a remedy: to induce “far-sightedness” by strengthening attraction at large distances, so that even distant nearest neighbors continue to be drawn together. We therefore augment UMAP’s attraction shape with a term that reinforces the attractive force at large ζ ,

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β regulates the strength of the added term (we use $\beta = 0.2$ throughout; Fig. 2(e)). At the level of the loss, modifying the attraction shape in this way is exactly equivalent to appending a norm-based regularizer to the attractive term, $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} (\beta/3) \|y_i - y_j\|_2^3$, so the modification carries a precise interpretation in either formulation. Beyond drawing distant pairs together, the added attraction promotes an early intermixing of points that aids convergence under random initialization, in close analogy with early exaggeration in *t*-SNE. We adopt the linear correction in (13) precisely because it is the simplest choice that achieves this effect; other monotone functions, such as $\log \zeta$ or ζ^p with $p \in \mathbb{R}_{\geq 0}$, would serve equally well.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

which reverts to the original UMAP shape after the first 100 epochs, thereby retaining the convergence benefit that f_a^M confers during the early, intermixing phase while removing any long-range distortion it might otherwise impose once the coarse layout is established. We now examine the effect of both the modified and the composite attraction shapes on dimensionality reduction from random initialization.

We first constructed a PCA-initialized embedding of MNIST (Fig. 2(a)) to serve as a reference, then generated embeddings from Gaussian random initialization for each attraction shape, repeating the procedure 100 times per shape. To quantify agreement, we use Procrustes analysis (Gower, 1975), which aligns two point clouds under translation, scaling, rotation, and reflection (Appendix D.1 gives the formal definition); we align each randomly initialized embedding to the PCA reference and take the residual separation as the Procrustes distance (p_d). Figure 2(b) presents the four default-UMAP embeddings with the lowest p_d : the individual cluster shapes are consistent, but their relative placements are not. The modified and composite attraction shapes (Figs. 2(c) and (d)) markedly improve the consistency of cluster placement.

To characterize placement consistency across all pairs of runs rather than a favorable few, we introduce the Procrustes matrix, whose diagonal is sorted by p_d relative to the PCA-initialized mapping and whose off-diagonal entries record the pairwise p_d between randomly initialized mappings (Appendix D.1), a construction analogous to the self-similarity matrix (Foote, 1999). Aggregated over 100 runs, the default UMAP embeddings bear little resemblance to one another (Fig. 2(f), mean $p_d = 0.78$), whereas both the modified (Fig. 2(g), 0.49) and composite (Fig. 2(h), 0.50) shapes yield embeddings that are substantially more mutually similar. Because the strengthened attraction alters only how the embedding is reached—not the graph it optimizes—the reduced p_d is naturally read as evidence that UMAP encodes an underlying structure largely independent of initialization: the residual variability that default UMAP exhibits reflects not what UMAP represents but the difficulty of reaching that representation in the low dimension, a difficulty that weak attraction at long distances imposes and that the modified shape substantially relieves.

5.2 Cluster Formation and Compactness

The parameter nominally intended to control cluster formation in UMAP is the minimum-distance parameter m_d (via Eq. 2). Varying m_d , however, simultaneously constrains the admissible values of a and b (Fig. 3(a)), so that reducing it produces embeddings with little discernible additional benefit (Figs. 3(b-d)); the underlying reason is that m_d exerts only a marginal influence on the repulsion shape (Fig. 3(e)). Varying a and b explicitly is the more direct route.

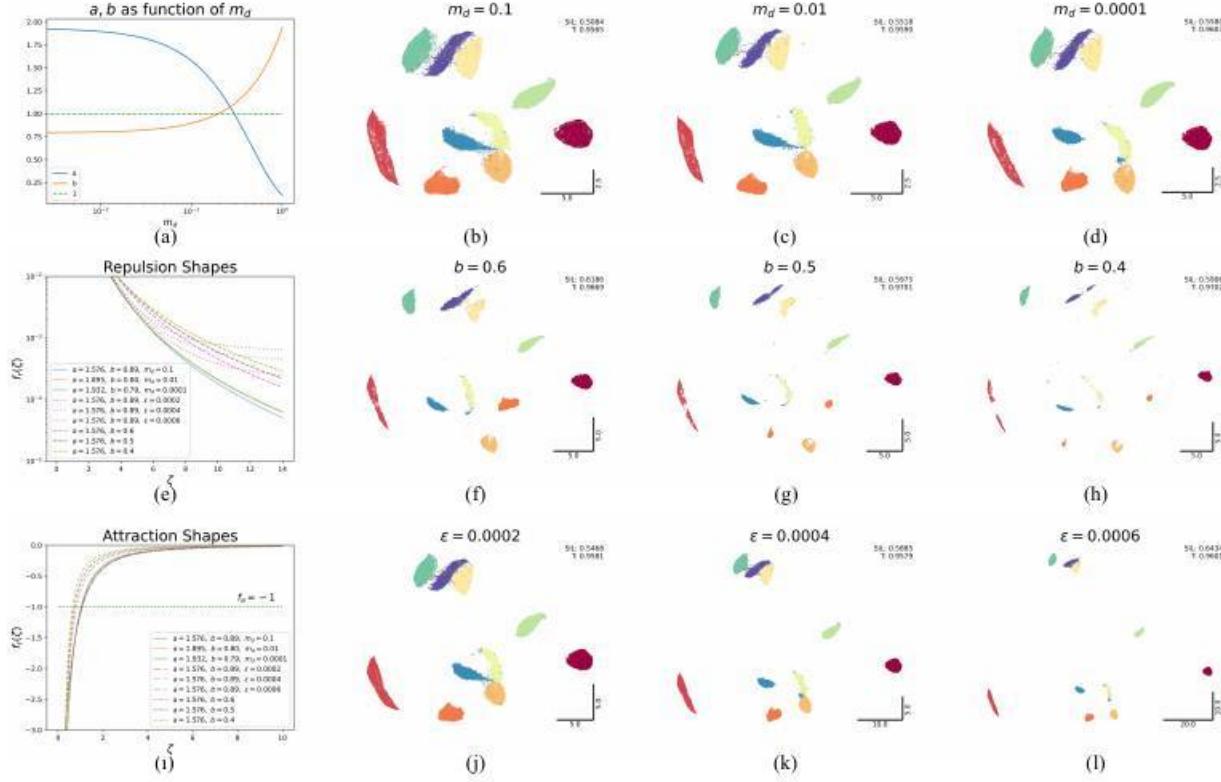


Figure 3: Control of inter-cluster distances on the MNIST dataset. (a) Deriving a and b by varying the minimum low-dimensional distance m_d traces out only a constrained region of the (a, b) parameter space. (b-d) The corresponding UMAP outputs, for m_d set to 0.1, 0.01, and 0.0001, show little gain in cluster compactness. (e) Repulsion shapes across parameter settings. (f-h) Increasing repulsion by explicitly decreasing b yields progressively more compact clusters and resolves clusters that were previously merged. (i) Attraction shapes across the same settings. (j-l) Increasing repulsion instead by adding a small positive constant ϵ to the repulsion shape widens inter-cluster distances while leaving attraction untouched.

Decreasing a does increase repulsion, but it attenuates attraction faster still and thereby aggravates near-sightedness; decreasing b affords far finer control (Figs. 3(e,i)). Lowering b to 0.6, 0.5, and 0.4 progressively widens inter-cluster distances and splits previously merged clusters (Figs. 3(f-h)). This splitting follows from the low-dimensional kernel growing increasingly heavy-tailed as b decreases, and heavy-tailed kernels are known to induce smaller, more distinct clusters (Van der Maaten and Hinton, 2008; Kobak et al., 2019) (see Kobak et al. (2019) for how b modulates the UMAP tail). The approach separates all ten MNIST labels into distinct clusters. It does not, however, isolate the mechanism: varying b amplifies repulsion more strongly than varying m_d does while simultaneously reshaping the attraction profile, so the individual contributions of attraction and repulsion remain entangled—a confound we disentangle next.

To isolate the effect of repulsion within this optimization, we hold attraction fixed and perturb repulsion alone, adding a small positive constant ϵ to the repulsion shape,

$$f_r^M = f_r^U + \epsilon, \quad (15)$$

with a and b held fixed. This is equivalent to appending a norm-based regularizer to the repulsive term of the loss, $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} (\epsilon/2) \|y_i - y_j\|_2^2$, and it produces repulsion stronger than any previously accessible through a , b , or m_d (Fig. 3(e)). Figures 3(j-l) confirm that inter-cluster distances grow with ϵ . The clustering behavior, however, mirrors that obtained by varying m_d : the labels separate only loosely as ϵ increases, and no new clusters form. Since ϵ leaves the attraction shape entirely untouched, and since varying m_d traces out essentially the same family of attraction shapes (Fig. 3(i)), the natural reading of these two controls is that, within UMAP’s optimization, cluster formation is driven predominantly by attraction rather than repulsion.

To sharpen this attribution, we vary the attraction and repulsion shapes one at a time, setting $b = 0.4$ in one shape while holding the other at its default (Fig. 4). With repulsion increased alone ($b = 0.4$), the default attraction produces no new structure and no new clusters, only more compact versions of the original UMAP clusters (Fig. 4(a)). With

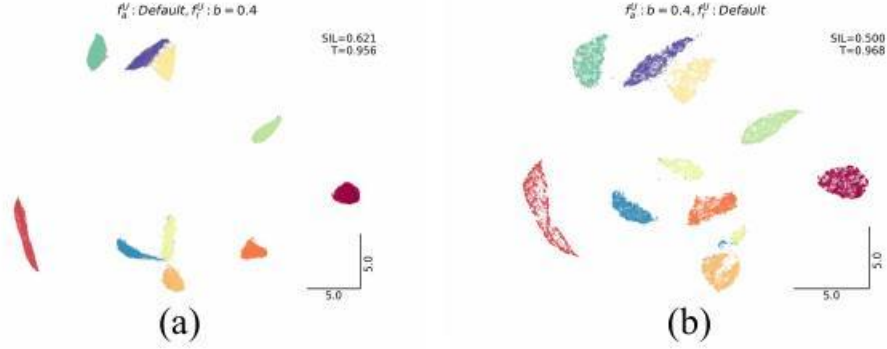


Figure 4: Embedding of the MNIST dataset under selectively modified shapes. (a) Default attraction shape combined with a repulsion shape at $b = 0.4$: the clusters coincide with those of default UMAP but are markedly more compact, so stronger repulsion alone tightens without restructuring. (b) Default repulsion shape combined with an attraction shape at $b = 0.4$: new substructure emerges within clusters and additional clusters form, so stronger attraction alone drives new structure. Varying one shape at a time isolates the distinct roles of attraction and repulsion.

attraction increased alone ($b = 0.4$) against the default repulsion, the embedding instead develops additional intra-cluster structure and several existing clusters split into smaller ones (Fig. 4(b)). This dissociation, obtained under the same graph, sampling scheme, and optimizer, points to a clear division of labor: attraction drives cluster formation, whereas repulsion controls compactness—tightening clusters or widening the gaps between them according to its shape. In Appendix G.3 we show that LocalMAP embeddings exploit precisely this attraction–repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

Fixing $a = b = 1$ is a common convention across dimensionality reduction algorithms, since it renders the gradients of many methods bounded and thereby eases optimization. Damrich et al. (2023) pursued this setting for UMAP and proposed NEG- t -SNE, which not only stabilizes the gradient but also yields more compact clustering even at a constant learning rate of 1. Parametric UMAP (Sainburg et al. 2021), meanwhile, initially adopted the original UMAP loss but subsequently replaced it with a numerically stable modified cross-entropy loss (Shi et al. 2023) built on a log-sigmoid kernel. Analyzing the two through their shapes reveals that they arrive at an identical formulation from different starting points—an equivalence we now state precisely.

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

With $a = 1$ and $b = 1$, the attraction and repulsion shapes of UMAP reduce to $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. The attraction shape is now bounded within $[-2, 0]$, attaining $f_a^U(0) = -2$ (Fig. 1(f)), whereas the repulsion shape is essentially unchanged and remains unbounded as $\zeta \rightarrow 0$ (Fig. 1(g)). Because $f_a^U < -1$ as $\zeta \rightarrow 0$, Proposition 4.1 and the analysis of Section 5 together imply that even this unit-parameter case retains a positive ζ_{-1} and therefore still requires learning-rate annealing.

Damrich et al. (2023) analyzed UMAP’s negative-sampling loss through the lens of contrastive embedding (CE) and identified the effective kernel of UMAP as $1/\zeta^2$. Within the CE framework, they introduced NEG- t -SNE by replacing this kernel with $1/(1 + \zeta^2)$; translated back into the UMAP formalism, this corresponds to the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$ and hence to the attraction and repulsion shapes $f_a^N = -2/(2 + \zeta^2)$ and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$ (Figs. 1(f,g)). The attraction shape is now bounded within $[-1, 0]$ and satisfies Proposition 4.1 so that every $\lambda \in [0, 1]$ induces contraction and the oscillation between expansion and contraction is eliminated. NEG- t -SNE is accordingly far less sensitive to learning-rate annealing, and its clusters remain sharp even at a constant $\lambda = 1.0$. Its repulsion shape is likewise bounded, within $[0, 1]$, and does not diverge as $\zeta \rightarrow 0$, so that fewer points are ejected from clusters under random sampling. Where Damrich et al. (2023) attributed this disparity solely to the bounded repulsive forces of NEG- t -SNE versus the unbounded ones of UMAP, the shape analysis of Fig. 1 identifies the bounded attraction shape as an equally consequential factor in cluster stability and compactness—a mechanism the repulsion-only account leaves out. Appendix F develops this comparison further with illustrations. Read together with the dissociation of Fig. 4 this finding carries a constructive implication: because the two shapes act independently, attraction and repulsion

shapes drawn from different methods can be recombined at will—pairing, for instance, UMAP’s attraction shape with NEG- t -SNE’s repulsion shape—a possibility we substantiate for UMAP, NEG- t -SNE, and PaCMAP in Appendix J.

7 Discussion and Conclusion

In this work, we investigated how attractive and repulsive forces give rise to cluster formation. Although it is widely accepted that attraction draws similar points together while repulsion drives dissimilar points apart, the precise mechanisms of these forces had not previously been studied in depth. By decomposing the forces into their shapes, we have exposed much of the dynamics that underlies cluster formation. The main text concentrates on UMAP and NEG- t -SNE, but the shape formalism itself is general: it applies to any attraction–repulsion method, as the treatment of Parametric UMAP, PaCMAP, LocalMAP, TriMap, t -SNE, SNE, and MDS in Appendix G demonstrates.

Treating the attraction and repulsion coefficients as functions and analyzing them through their respective shapes (Propositions 4.1 and 4.2) uncovered a genuinely counterintuitive result: for sufficiently close pairs, UMAP’s attractive term expands rather than contracts the distance (Fig. 1(b)), pushing neighboring points apart. We traced the standard remedy for this behavior to UMAP’s learning-rate schedule, which drives the minimum distance for contraction ζ_{-1} toward zero.

We further examined the initialization of the low-dimensional embedding. Nearest neighbors that begin far apart experience attraction too weak to contract them within the optimization budget; the formalism supplies a principled correction—augmenting the attraction shape to strengthen attraction at large distances. Across MNIST, Fashion-MNIST, and single-cell transcriptomes, this modification reduced the mean Procrustes distance to a PCA-initialized reference by roughly a third to a half, evidence that UMAP encodes an underlying structure largely independent of initialization and that the customary numerical tricks chiefly serve to reach it. These gains are demonstrated for the datasets and consistency metric studied here; whether strengthened attraction also benefits downstream tasks such as classification, clustering, or retrieval is a question our experiments do not resolve and that we leave to future work.

Analysis of the repulsion shape showed that stronger repulsion widens inter-cluster distances. Exploiting this, we added a small positive constant to the shape and reached levels of repulsion inaccessible through the conventional route of tuning the minimum-distance parameter m_d .

Finally, we clarified the relationship between NEG- t -SNE and UMAP. At $a = b = 1$, the decisive difference is that NEG- t -SNE, in its default form, has bounded attraction ($\in [-1, 0]$) and bounded repulsion ($\in [0, 1]$), so that both propositions hold irrespective of the learning rate and the method is correspondingly insensitive to any fixed $\lambda \in [0, 1]$. Once a and b are allowed to vary, however, NEG- t -SNE inherits exactly the numerical challenges of UMAP. We additionally proved that NEG- t -SNE and Parametric UMAP are formally equivalent.

We note that the propositions certify the behavior of individual updates, and the empirical claims are grounded in the datasets, metrics, and settings examined here; extending the local analysis to a full account of the global stochastic dynamics under many simultaneous samples remains open. Even so, the shape formalism furnishes concrete tools for interpreting and controlling dimensionality reduction, and the close connection between dimensionality reduction and contrastive learning (Damrich and Hamprecht, 2021; Hu et al., 2023) suggests that these tools may extend naturally to representation learning—a direction we regard as especially promising for future research.

Software and Data

All datasets used in this paper are publicly available. The MNIST dataset is available at yann.lecun.com. Fashion-MNIST is available at github.com. Single-cell transcriptomes data is available at github.com.

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Appendix

The appendices develop the supporting material in the following order. Appendix [A](#) provides the derivations of UMAP’s attraction and repulsion shapes and the proofs of the associated propositions. Extending the main-text discussion, Appendix [B](#) revisits the constant-learning-rate regime, and Appendix [C](#) demonstrates the implications of strengthened attraction and repulsion for large language model embeddings. Appendix [D.1](#) formally defines the Procrustes distance and Procrustes matrix, and Appendix [E](#) reproduces the random-initialization experiment on additional datasets. Appendix [F](#) extends the comparison of NEG-*t*-SNE, Parametric UMAP, and UMAP, after which Appendix [G](#) analyzes a broad range of alternative dimensionality reduction methods within the same shape formalism. Appendix [H](#) details the construction of the high-dimensional graph across these methods. Appendix [I](#) reports the detailed results of the λ_a - λ_r sweep of Fig. [1](#) for UMAP, NEG-*t*-SNE, and PaCMAP across datasets, and Appendix [J](#) presents the effect of mixing and matching shapes drawn from different algorithms. Appendix [K](#) closes with full implementation details.

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. We subtract Eq. [\(7\)](#) from Eq. [\(8\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. We subtract Eq. [\(9\)](#) from Eq. [\(10\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

To derive the attraction shape in full generality, we adopt the general low-dimensional affinity $q_{ij} = (\gamma + a\|y_i - y_j\|_1^{2b})^{-1}$, which subsumes both methods of interest: it recovers UMAP for $\gamma = 1$ and NEG-*t*-SNE for $\gamma = 2$. The attractive force is then

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

With $\zeta = \|y_i - y_j\|_2$, the leading scalar factor constitutes the attraction shape,

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

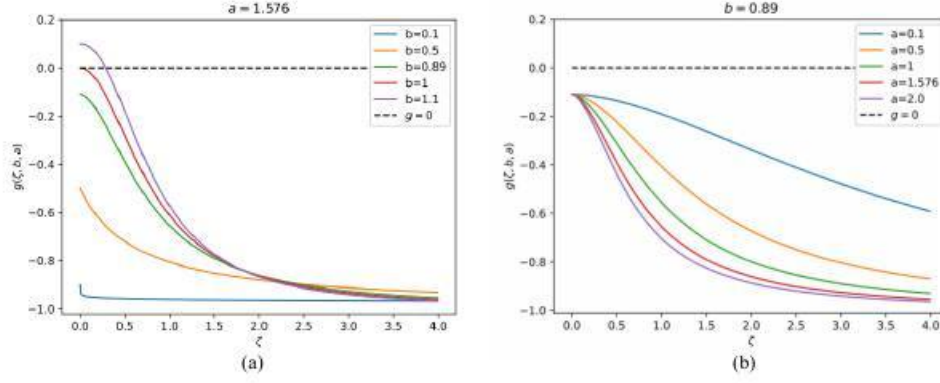


Figure 5: Values of $g(\zeta, b, a)$ with (a) a fixed at 1.576 and (b) b fixed at 0.89. The attraction shape f_a^U is strictly increasing wherever $g < 0$, a condition that holds throughout the range $0 < b \leq 1$ assumed in the analysis.

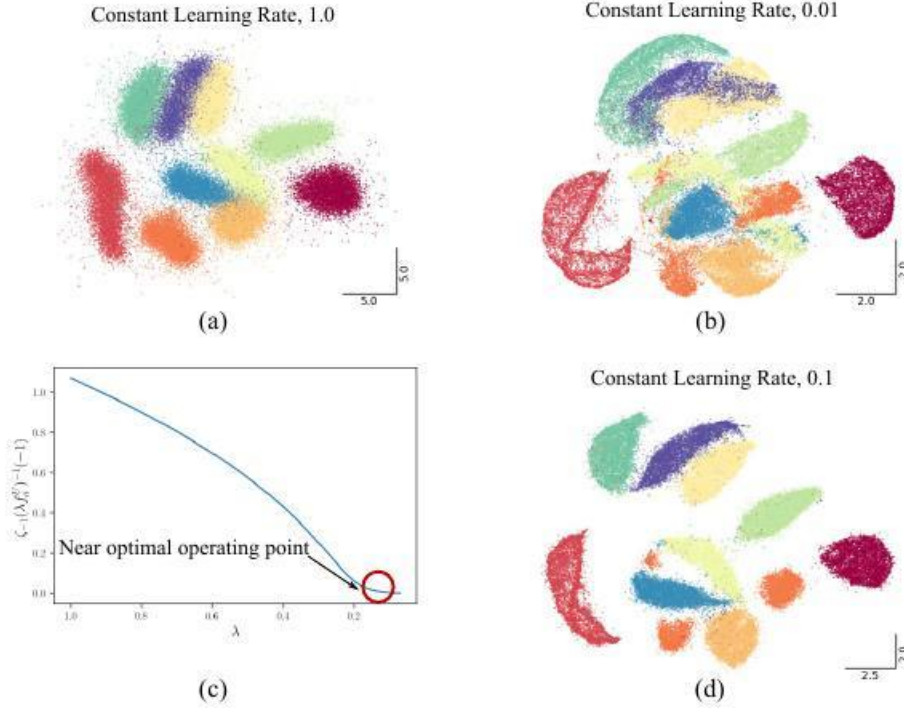


Figure 6: Effect of a constant learning rate on the embeddings. (a) When λ is too large ($\lambda = 1.0$), the resulting large ζ_{-1} leaves the embedding diffuse. (b) When λ is too small ($\lambda = 0.01$), attraction and repulsion are both too weak and no clusters form. (c) ζ_{-1} decreases nonlinearly as λ decreases; the operative objective is to drive ζ_{-1} toward zero while preserving effective attraction and repulsion. (d) At the near-optimal constant learning rate $\lambda = 0.1$, distinct and compact clusters emerge.

A.4 Condition for strictly increasing f_a^U :

The dependence of f_a^U on distance is governed by its derivative,

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

which yields the strictly increasing condition ($\frac{df_a^U}{d\zeta} > 0$)

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. Examining the derivative and asymptotes of g establishes that this inequality holds precisely when $0 < b \leq 1$. Figure 5 plots g across values of b and a : once b exceeds 1, inequality (23) fails and f_a^U ceases to be monotone, which is why the main-text analysis restricts attention to $b \leq 1$.

A.5 Repulsion Shape:

Under the same general low-dimensional affinity, the repulsive force follows by an analogous computation,

$$\begin{aligned} \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\ &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\ &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j). \end{aligned} \quad (24)$$

The leading scalar factor is the repulsion shape,

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}, \quad (25)$$

which is positive throughout, $f_r > 0$, so that Proposition 4.2 is satisfied unconditionally.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function of the attractive term under the modification in Eq. (13) is

$$-\int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \quad (26)$$

whereas the repulsive term under Eq. (15) is

$$-\int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \quad (27)$$

In both cases the additional term acts as a norm-based regularizer of the embedding coordinates. Introducing the modification directly at the level of the shapes, rather than the loss, carries a definite interpretive advantage: for the attractive term the added component increases far-sightedness, whereas for the repulsive term it contributes a constant repulsive coefficient—readings that the equivalent loss-level regularizers obscure.

B More on UMAP’s Learning Rate

As discussed in Sections 5 and 6, UMAP is widely held to require learning-rate annealing (Fig 6). Section 5 accounted for this requirement by introducing the minimum distance for contraction (ζ_{-1}) and establishing that annealing, by reducing ζ_{-1} , produces compact clusters. Section 6 then compared the attraction shapes of UMAP (at $a = 1.0$, $b = 1.0$) and NEG- t -SNE, showing that NEG- t -SNE tolerates a constant learning rate of 1.0 better than UMAP because its attraction shape lies within $[-1, 0]$ whereas UMAP’s lies within $[-2, 0]$. The same reasoning predicts that UMAP itself can withstand a constant learning rate of 0.5, since halving λ confines its effective attraction shape to $[-1, 0]$ (Figs. 11(f,h)).

Even so, across a wide range of a and b , annealed embeddings remain superior, because the ultimate objective is to drive ζ_{-1} to zero—which occurs only as the learning rate approaches zero—and short of that limit the embedding stays diffuse (Fig 6(a)). Setting the learning rate too low from the outset creates the opposite failure: attraction and repulsion are then both too weak for clear clusters to form (Fig 6(b)). The ζ_{-1} -versus- λ curve (Fig 6(c)) locates a near-optimal operating point at $\lambda = 0.1$, where ζ_{-1} is already small while attraction and repulsion retain substantial strength; at this constant learning rate we indeed recover compact clusters with clean boundaries (Fig 6(d)).

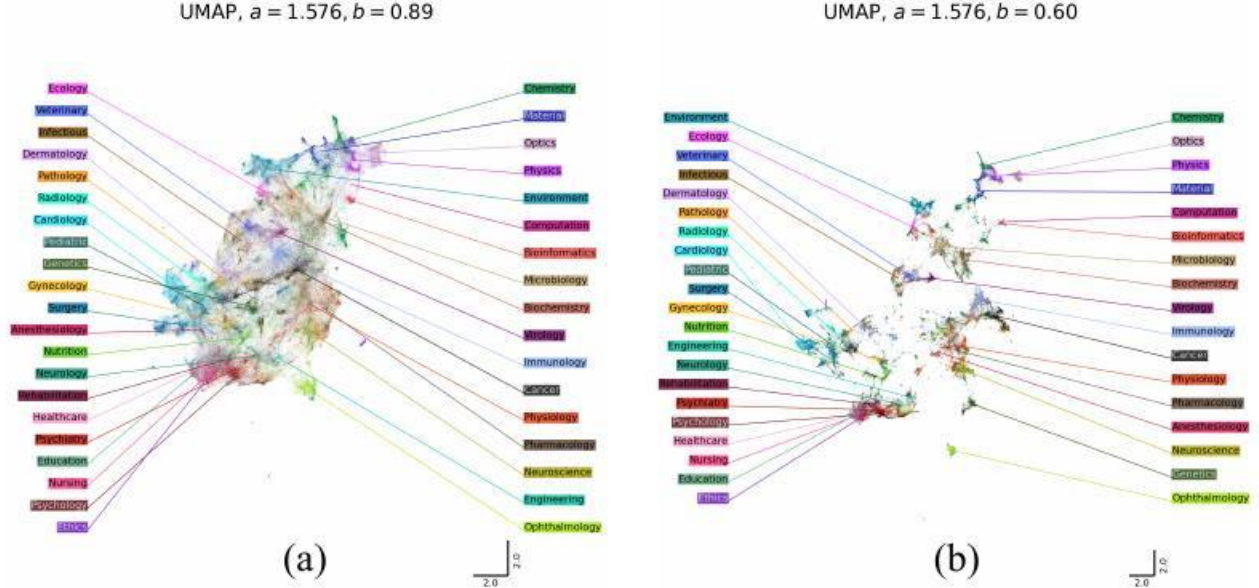


Figure 7: Visualization of 300K PubMed abstracts embedded from PubMedBERT features. (a) Default UMAP ($a = 1.576, b = 0.89$) yields a crowded mapping in which the label colors overlap heavily, whereas (b) UMAP with strengthened repulsion and attraction ($a = 1.576, b = 0.60$) produces well-formed, non-overlapping label clusters. Both mappings are annotated for readability.

C Large Language Model Embedding

The benefit of strengthening attraction and repulsion together extends beyond image data to high-dimensional language model representations. To demonstrate this, we drew 300,000 PubMedBERT (Gu et al., 2021) features with known labels from the PubMed dataset (González-Márquez et al., 2024). Default UMAP yields a crowded structure in which the label colors overlap (Fig. 7(a)); stripped of the labels, the embedding would be difficult to interpret, as no explicit clusters are visible. Reducing b to 0.6—the prescription of Section 5.2—increases repulsion while, through the accompanying gain in attraction, promoting cluster formation, and the mapping becomes readily explorable even without labels (Fig. 7(b)). The improvement is quantitative as well as visual: on a held-out set of 10,000 points, k -NN accuracy ($k = 10$) rises from 49.2% to 55.48% when b is reduced to 0.6.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower [1975]) quantifies the similarity between two point clouds $\{x\}$ and $\{y\}$ up to the linear transformations of translation, scaling, and rotation. Operationally, one holds $\{x\}$ fixed and transforms $\{y\}$ until the two sets are maximally aligned. Writing $\{y'\}$ for the optimally transformed $\{y\}$, the Procrustes distance is

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

As an alignment-invariant measure, the Procrustes distance has proven valuable across a variety of settings (McInnes et al., 2018; Islam and Fleischer, 2022; Kotlov et al., 2024), and it is well suited to comparing embeddings that are meaningful only up to rotation, reflection, and scale.

We use it here to quantify the consistency of embeddings under random initialization. Let $\{x\}_p$ denote a reference embedding (PCA-initialized in our experiments) and $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ a set of N embeddings obtained from random initialization. We summarize overall similarity by the mean and standard deviation of the strictly

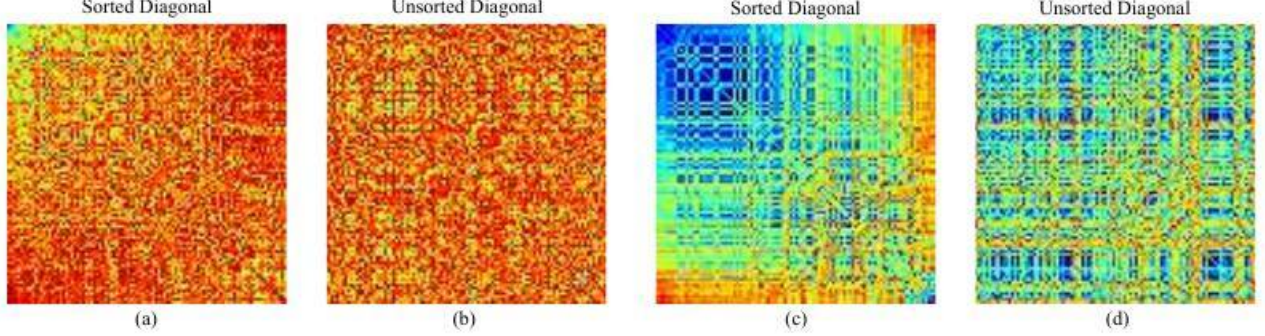


Figure 8: Effect of sorting the diagonal of the Procrustes matrix on its visualization. (a) Procrustes matrix reproduced from Fig. 2(f) for the default UMAP attraction shape, with the diagonal sorted by p_d relative to a PCA-initialized reference embedding. (b) The same data with the diagonal left unsorted. (c) Procrustes matrix reproduced from Fig. 2(g) for the modified attraction shape, again sorted by p_d relative to the PCA-initialized reference. (d) The same data with the diagonal left unsorted. Sorting is a purely visual reordering—the entries are unchanged—yet it exposes the self-similar block that an arbitrary ordering conceals.

lower-triangular entries of the matrix P (reported as mean $\pm$ std in Figs. 2(9) and 10),

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indices of X_r can be ordered so that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, whereupon the diagonal entries of the Procrustes matrix are

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal entries are

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Sorting the diagonal changes nothing numerically, but visually it exposes the underlying self-similarity of the embeddings. In Fig. 8(a), for instance, the sorted diagonal reveals that mutually similar embeddings concentrate in the upper-left region of the matrix. Under the modified attraction shape the population of mutually similar embeddings grows substantially—witness the enlarged blue region in Fig. 8(c)—consistent with the algorithm settling into a metastable configuration. Left unsorted, this block structure dissolves and the self-similarity becomes invisible (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski 2001) quantifies the extent to which local neighborhoods are preserved through dimensionality reduction,

$$T = 1 - \frac{2}{nk(2n-3k-1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. We fix $k = 5$ throughout, thereby assessing preservation of each point’s five nearest neighbors. For computational efficiency we evaluate trustworthiness on 10,000 randomly sampled indices, holding the same indices fixed across every pair of embeddings being compared so that the metric remains directly comparable.

D.3 Silhouette Score

The silhouette score (Rousseeuw 1987), though designed to assess clustering algorithms, we repurpose to evaluate label separation in the embeddings—how faithfully the ground-truth labels are grouped—on the premise that a good embedding should render the labels as sharply separated as the data permit. For a point y_i in a point cloud $\{y\}$, let a_i be

the mean distance from y_i to the remaining points of its own label and b_i the minimal mean distance from y_i to the points of any other label. The pointwise silhouette is then

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies within $[-1, 1]$: values near 1 indicate that y_i sits well within its own label cluster, values near 0 mark boundary points, and values near -1 signal that the labels have failed to separate. The overall silhouette score is the mean over all points,

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We compute the silhouette score over the entire embedding, without random sampling, under the Euclidean metric.

E Effect of Random Initialization in Additional Datasets

The main text (Section 5.1) reported results on MNIST. Here we replicate the identical experiment—100 random initializations per shape, aligned by Procrustes analysis to a PCA-initialized reference—on the Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) datasets (Figs. 9 and 10). The central conclusion holds across all three datasets: modified and composite attraction shapes—those that strengthen attraction at large distances—markedly improve reconstruction consistency, reducing the mean Procrustes distance from 0.72 to 0.38 on FMNIST and from 0.91 to 0.61 on the transcriptomes data. The consistency of the effect across an image benchmark, a fashion-image benchmark, and a biological single-cell dataset indicates that it is a property of the attraction shape rather than of any one data modality.

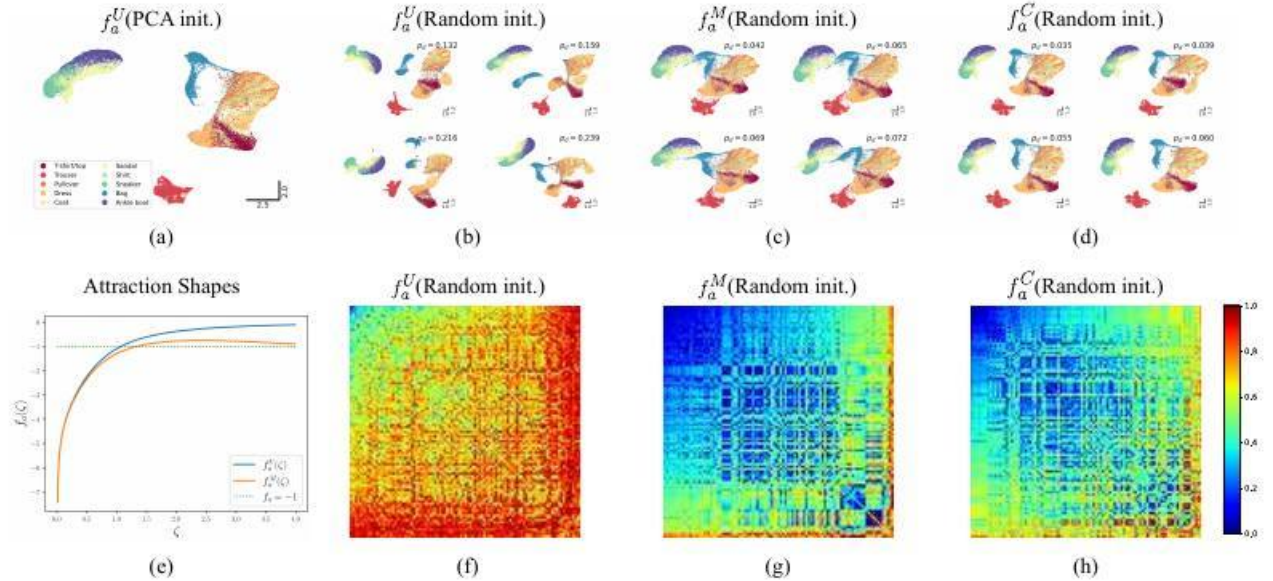


Figure 9: Effect of random UMAP initialization on different attraction shapes for the FMNIST data. (a) Reference mapping obtained with PCA initialization. (b-d) The four mappings with the lowest Procrustes distance (p_d) from the reference (a), for the (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices aggregated over 100 runs for the (f) default ($p_d = 0.72 \pm 0.15$), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes; the modified and composite shapes roughly halve the mean p_d , replicating the MNIST result of Fig. 2.

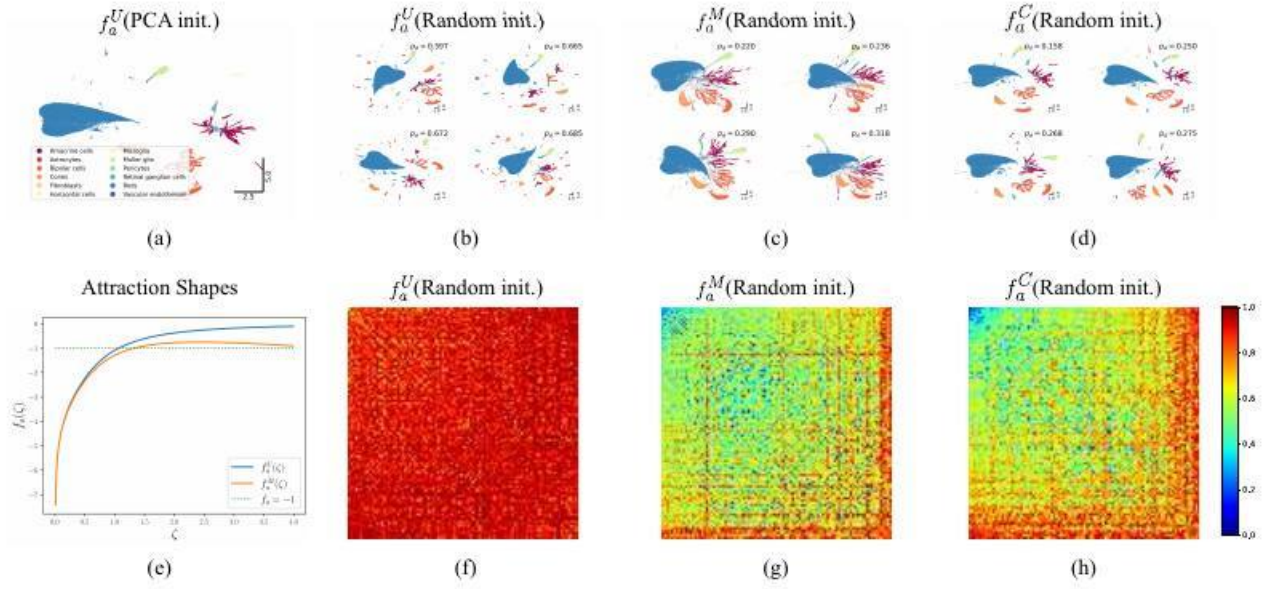


Figure 10: Effect of random UMAP initialization on different attraction shapes for the single-cell transcriptomes data. (a) Reference mapping obtained with PCA initialization. (b-d) The four mappings with the lowest Procrustes distance (p_d) from the reference (a), for the (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices aggregated over 100 runs for the (f) default ($p_d = 0.91 \pm 0.06$), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes; the modified shape reduces the mean p_d substantially, extending the effect to a biological dataset.

F More on NEG-*t*-SNE

F.1 Comparison to UMAP

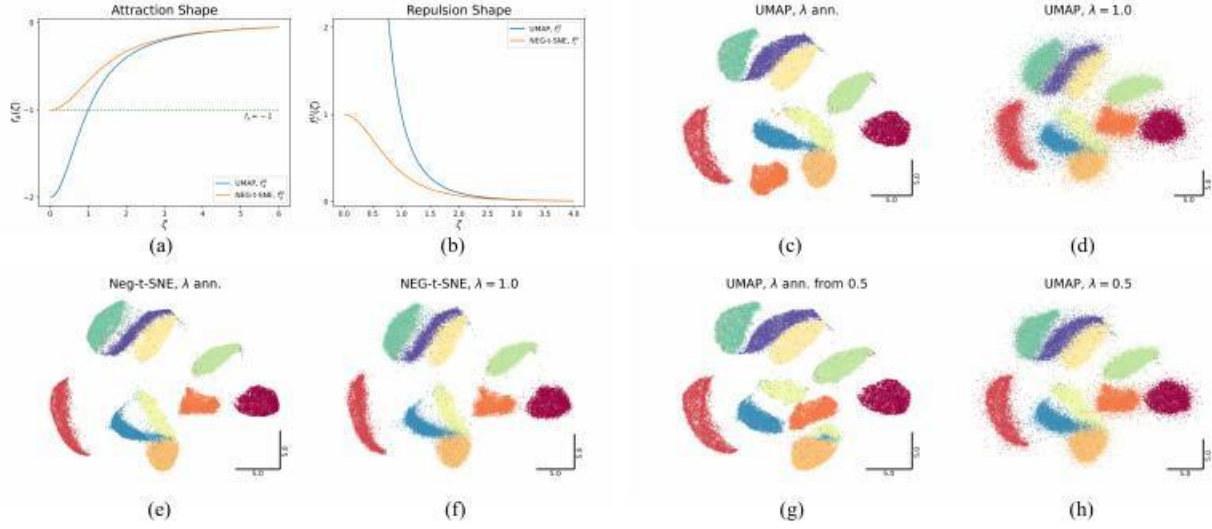


Figure 11: Sensitivity of UMAP and NEG-*t*-SNE to the learning rate on the MNIST dataset. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1, b = 1$) and NEG-*t*-SNE. (c,d) UMAP is highly sensitive to the learning rate λ : because $f_a^U < -1$ as the separation ζ decreases, its clusters become fuzzy without annealing. (e,f) NEG-*t*-SNE is far less sensitive to λ , since $f_a^N \in [-1, 0]$ for every ζ , and its clusters stay sharp even without annealing. (g,h) Confining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ likewise reduces UMAP’s sensitivity to λ , as the analysis predicts.

Figure 11 presents the shapes of UMAP and NEG-*t*-SNE alongside a range of MNIST embeddings. Under annealed learning rates the two methods produce similar output (Figs. 11(c,e)); at a constant $\lambda = 1$, however, UMAP yields a fuzzy structure whereas NEG-*t*-SNE retains much cleaner boundaries (Fig. 11(d,f)). The analysis of Section 6 makes a testable prediction: confining f_a^U to $[-1, 0]$ should sharpen UMAP’s clusters at fixed λ —a prediction already foreshadowed in Fig. 1(h,i), where UMAP embedded and clustered better for an attraction learning rate $\lambda_a \lesssim 0.5$. The simplest realization sets $\lambda = 0.5$, which satisfies Proposition 4.1 for all ζ . The resulting embeddings (Figs. 11(g) and (h)) bear out the prediction: the clusters closely resemble those of NEG-*t*-SNE and, at constant $\lambda = 0.5$, are markedly sharper than before (Fig. 11(d,h)). A handful of points still lie outside the clusters—a residual effect of UMAP’s unbounded repulsion shape that NEG-*t*-SNE, with its bounded repulsion, resolves.

Introducing the parameters a and b into NEG-*t*-SNE recovers precisely the Parametric UMAP formulation (Proposition 6.1). The affinity becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, with attraction and repulsion shapes $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2 + a\zeta^{2b}}$ and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1 + a\zeta^{2b})(2 + a\zeta^{2b})}$. For $0 < b < 1$ both shapes diverge as $\zeta \rightarrow 0$, so once a and b are free to vary NEG-*t*-SNE inherits the numerical challenges of UMAP together with the attendant limitations. The two methods do differ in one respect: NEG-*t*-SNE’s attraction shape attains a smaller minimum distance for contraction ζ_{-1} than UMAP’s, which can aid cluster formation, but its more rapid decay to zero intensifies near-sightedness and may weaken attraction over longer distances—a trade-off rather than an unqualified advantage.

F.2 Comparison to Parametric UMAP

Parametric UMAP was originally trained with the UMAP objective (Sainburg et al. 2021), but subsequent work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al. 2023). Under this loss Parametric UMAP and NEG-*t*-SNE (Damrich et al. 2023) coincide (Proposition 6.1), as we now establish by direct computation.

Proof of Proposition 6.1

Proof. Under this modification the kernel function becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term evaluates to

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

and the repulsive term to

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both expressions coincide with those of NEG- t -SNE carrying explicit parameters a and b (whereas NEG- t -SNE fixes both to 1), which establishes the equivalence. $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms considered here all share UMAP’s kernel function (with $a = 1$ and $b = 1$ in their low-dimensional weight),

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We begin with TriMap (Amid and Warmuth [2019]). Although this algorithm operates on triplets rather than pairwise interactions, it serves as an instructive first step toward analyzing attraction and repulsion in settings more involved than UMAP. We then turn to Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al. [2021]) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al. [2025]), both of which recast TriMap’s triplet loss in terms of pairwise interactions. Next we treat t -SNE (Van der Maaten and Hinton [2008]), follow with a short note on SNE (Hinton and Roweis [2002]) and its alternative kernel, and close with multidimensional scaling (Borg and Groenen [2007]). That the same shape formalism accommodates each of these methods, despite their differing loss functions and sampling schemes, is direct evidence of its generality.

G.1 TriMap

TriMap (Amid and Warmuth [2019]) optimizes the low-dimensional embedding through a triplet loss,

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the weight of the triplet (y_i, y_j, y_k) , y_j belongs to the high-dimensional k -nearest-neighbor set of y_i , and y_k is a distant point. Minimizing this loss simultaneously draws y_i and y_j together and pushes y_i and y_k apart. The corresponding update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the nearest-neighbor distance, $\zeta_2 = \|y_i - y_k\|_2$ is the distance to the distant point, and f_a^T and f_r^T are the attraction and repulsion shapes of TriMap. In contrast to UMAP, the attractive and repulsive components

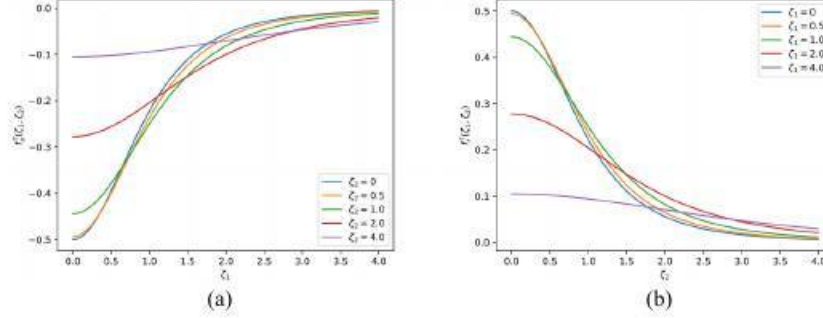


Figure 12: (a) Attraction shapes of TriMap across values of ζ_2 and (b) repulsion shapes of TriMap across values of ζ_1 , illustrating that each shape depends on both distance arguments—the feature that distinguishes TriMap’s two-dimensional shapes from UMAP’s one-dimensional ones.

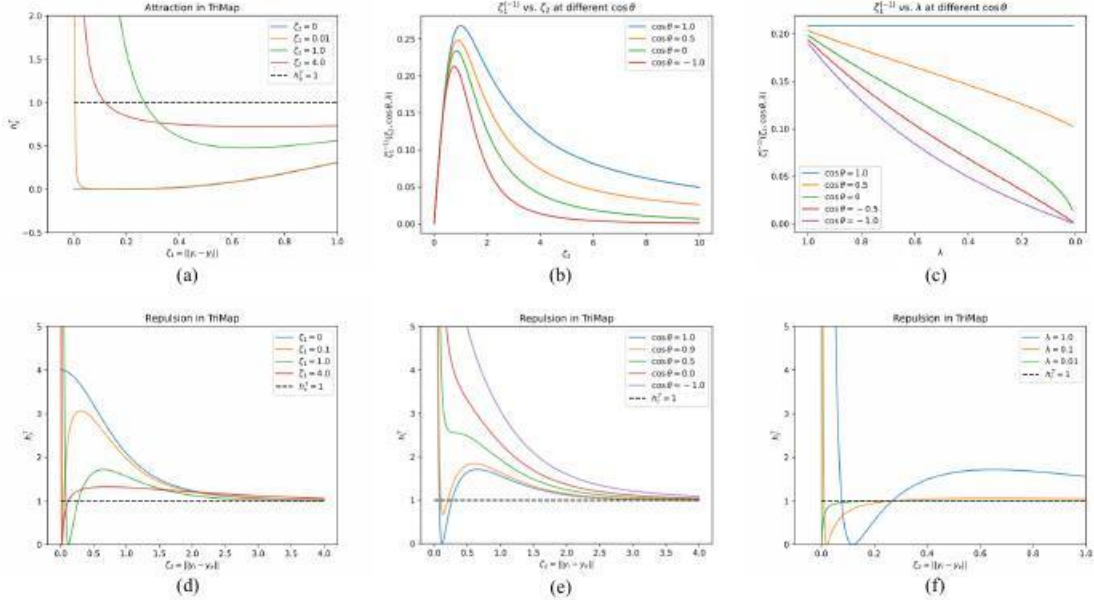


Figure 13: Attraction and repulsion behavior in TriMap. (a) h_a^T versus ζ_1 across values of ζ_2 ; values below the dotted line indicate attraction. As in UMAP, TriMap exhibits both attraction and repulsion for the nearest-neighbor pair (y_i, y_j) . (b,c) Unlike UMAP, the minimum distance for contraction $\zeta_1^{(-1)}$ is not fixed but varies with (b) ζ_2 and (c) λ , with $\cos \theta$ setting the range of these values. (d-f) Repulsion in TriMap as a function of (d) ζ_1 , (e) $\cos \theta$, and (f) λ ; values above (below) the dotted line indicate repulsion (attraction). Whereas UMAP’s repulsive force can only repel, TriMap’s can either attract or repel. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

do not separate, and each shape depends on both distances, rendering the shapes two-dimensional. The attraction shape takes the form

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

The attraction and repulsion shapes of TriMap (Fig. 12) follow trends similar to UMAP's. One key difference is that the attraction shape bottoms out at -0.5 , so that—unlike UMAP—attraction alone never flips the position of a pair. Because Eqs. (45)-(47) do not decouple the attractive and repulsive terms, however, Propositions 4.1 and 4.2 do not apply directly, and a coupled analysis is required. Taking attraction first, we establish the following.

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends jointly on ζ_1 , ζ_2 , $\cos \theta$, and λ ; in particular, the distance $\zeta_1 = \|y_i - y_j\|_2$ at which contraction is sought is now coupled to these additional variables rather than fixed. Figure 13(a) traces the attraction behavior across values of ζ_2 at $\cos \theta = 1$ and $\lambda = 1$: values below the dotted line indicate attraction, and hence contraction of ζ_1 , while values above indicate repulsion, and hence expansion. The intersection of h_a^T with the dotted line marks the minimum distance for contraction $\zeta_1^{(-1)}$ —the TriMap analogue of UMAP's ζ_{-1} —defined as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

In most cases $\zeta_1^{(-1)}$ is finite and strictly positive (Fig. 13(b,c)). TriMap therefore behaves much like UMAP and requires learning-rate annealing or an equivalent mechanism; in practice the TriMap implementation supplies this through the delta-bar-delta method (Jacobs 1988) together with an appropriate initialization.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) likewise depends on ζ_1 , ζ_2 , $\cos \theta$, and λ , but the quantity of interest is now the expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) chart the repulsion behavior as the remaining variables vary: values above the dotted line indicate repulsion, and hence expansion, while values below indicate attraction. The striking contrast with UMAP is that TriMap's repulsive term can itself produce contraction rather than expansion. Because this anomaly is confined to small distances, a suitable initialization and choice of triplets suffice to avoid it in practice.

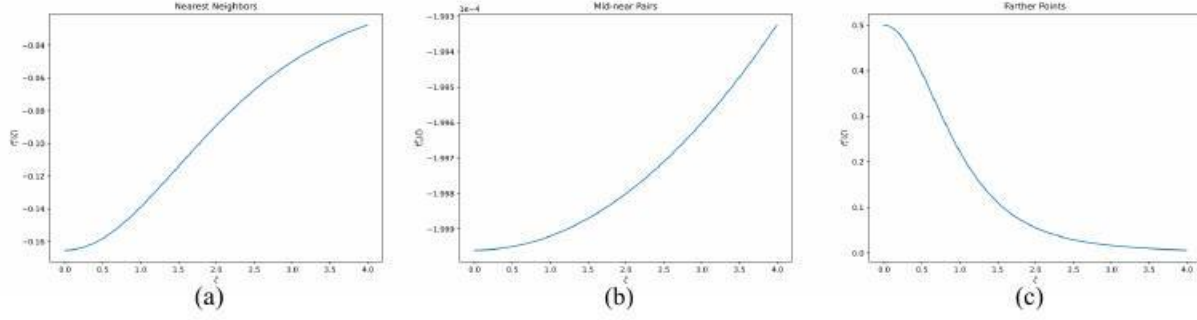


Figure 14: Attraction and repulsion shapes of PaCMAP. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near pairs; note that the mid-near values are on the order of 10^{-4} , so this term contributes negligibly to the dynamics. (c) Repulsion shape for the farther pairs. All panels use $\lambda = 1$.

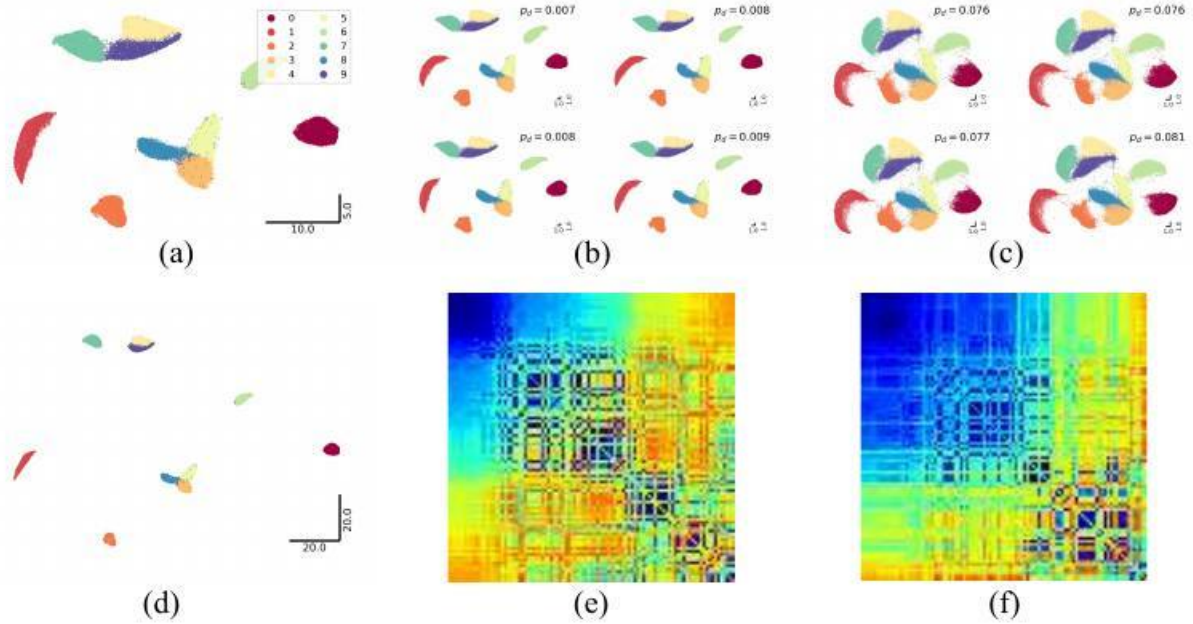


Figure 15: PaCMAP behavior under different conditions. (a) PaCMAP embedding of MNIST with PCA initialization. (b,c) The four randomly initialized samples that best match (a) under (b) original PaCMAP and (c) PaCMAP with the modified attraction shape $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP of MNIST with the modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP ($p_d = 0.51 \pm 0.22$) and (f) PaCMAP with the modified attraction shape (0.43 ± 0.22). As with UMAP, strengthening attraction at farther distances improves consistency, while strengthening repulsion yields smaller clusters and larger inter-cluster distances—evidence that the shape formalism, and the interventions it motivates, transfer directly to PaCMAP.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes the low-dimensional embedding across multiple scales at once. Its loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} weight the nearest-neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (see Appendix H). The first two terms supply attraction and the last supplies repulsion. The loss is, in effect, a modified TriMap triplet loss: in the attractive terms it substitutes constants for TriMap’s affinity to distant points ($1/10$ for nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther points). This substitution separates the

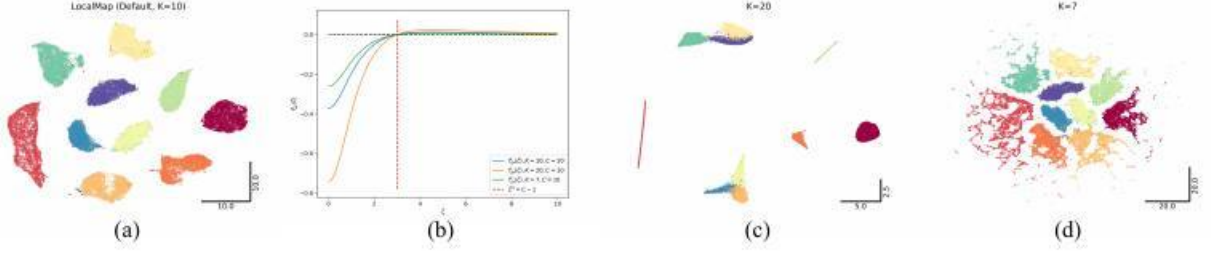


Figure 16: Behavior of LocalMAP on MNIST. (a) Default embedding, with all ten clusters cleanly separated. (b) Combined attraction–repulsion shape for nearest neighbors; values below (above) the dotted line indicate attraction (repulsion), and the transition from attraction to repulsion occurs as ζ increases past $\sqrt{C-1}$. Following the LocalMAP implementation, we set $C = 10$. (c) For large K ($= 20$), repulsion dominates and the clusters grow compact but the inter-cluster bridge persists. (d) For small K ($= 7$), attraction dominates and the clusters break apart.

loss into three additive terms and thereby decouples the update equations. The nearest-neighbor updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

the mid-near updates are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and the farthest-pair updates are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, and $\zeta_3 = \|y_i - y_l\|_2$ are the respective distances, f_a^P and f_m^P are the attraction shapes for nearest-neighbor and mid-near pairs, and f_r^P is the repulsion shape for the farthest pairs. The corresponding functional forms are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

Here f_a^P and f_m^P satisfy Proposition 4.1 and f_r^P satisfies Proposition 4.2 (Fig. 14). PaCMAP’s attraction is considerably weaker than UMAP’s yet remains effective across a wide range of learning rates, modulated by the Adam optimizer (Kingma and Ba, 2015); with $w_{NB} = 3$ the maximum attraction stays below 0.5 in magnitude, which precludes any flip during the attractive update. PaCMAP typically initializes the embedding within a small low-dimensional sphere—on the order of 10^{-3} in (Wang et al., 2021)—and relies on repulsion to drive the clusters apart. Overall it reproduces UMAP’s clustering behavior, especially on MNIST, with improved global ordering. Its consistency under random initialization already exceeds UMAP’s (Fig. 15(b,e)), and, exactly as the shape analysis predicts for UMAP, it improves further under a modified attraction shape (Fig. 15(c,f)). Adding a small constant to the repulsion shape strengthens repulsion and, by widening inter-cluster distances, makes the embedding more compact—the same intervention, with the same effect, that we introduced for UMAP.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al., 2025) refines the PaCMAP algorithm. Among its defining features is the clean separation of all ten MNIST clusters (Fig. 16(a)), a behavior that closely mirrors the effects we produced in Figs. 3(g,h,j,k) and Figs. 4(a,b). We now show how, through its shapes, the interplay of attractive and repulsive forces governs the compactness and connectedness of the resulting clusters.

LocalMAP first runs PaCMAP and then performs additional optimization on the attraction to decouple selected clusters, minimizing the loss

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term fuses the attractive and repulsive aspects of the triplet loss into a single function acting on one pair: in one distance regime it attracts, in the other it repels. The second term is identical to PaCMAP’s, save that the algorithm resamples the further pairs every few iterations. We therefore analyze the first, nearest-neighbor term alone. Its update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_j^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the combined attraction–repulsion shape,

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update contracts whenever $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ (Proposition 4.1); when $\zeta^2 > C-1$, we have $f_{ar} > 0$ and, by Proposition 4.2 the update expands. The parameters K and C jointly set the balance between attraction and repulsion. The LocalMAP implementation fixes both at 10 (Fig. 16b)), yielding attraction stronger than PaCMAP’s (Fig. 14a)). Consequently, for $\zeta > 3$ the nearest neighbors experience repulsion, so that pairs bridging two clusters are driven apart, and the scaling parameter K regulates the extent of this separation.

By Proposition 4.1 the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ yields the maximum K —and hence the maximum attraction attainable by f_{ar}^L —that leaves pair placements unflipped. Avoiding such flips during LocalMAP optimization is essential: they would disrupt the ordering inherited from PaCMAP and thereby impede cluster separation. The condition simplifies (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, giving $K \lesssim 13.44$ for $C = 10$ and $\lambda = 1$. Above this threshold ($K \gtrsim 13.44$) the repulsive forces dominate and the clusters grow more compact, yet the LocalMAP objective fails, as the inter-cluster bridge persists (Fig. 16c) at $K = 20$). As K decreases, conversely, the attractive forces dominate—repulsion having grown too weak—and existing clusters break apart (Fig. 16d) at $K = 7$, echoing Fig. 4b)). Both failure modes are precisely those the attraction–repulsion balance predicts.

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton 2008) optimizes pairwise distances through the loss

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the weight of the pair and, in the original implementation, t -SNE considers all pairs rather than nearest neighbors alone. This loss admits the same attractive–repulsive decomposition,

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term supplies the attractive forces and the second the repulsive forces. The attractive term is identical to UMAP’s (with $a = 1$, $b = 1$) and cheap to compute, whereas the repulsive term couples every pair through the normalizer and is therefore expensive. Applying the same decomposition used for UMAP yields the t -SNE update equations; because the original t -SNE forgoes the nearest-neighbor graph, the weight $w_{i,j}$ —computed over all pairs—enters the updates explicitly. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The repulsive updates are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$. These shapes take the functional forms

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1 + \zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1 + \zeta^2)^2}. \quad (77)$$

From the attractive updates (Eqs. 72-73), f_a^{t-SNE} satisfies Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and defines a minimum distance for contraction ζ_{-1} , exactly as in UMAP. The repulsive update, coupled across all pairs, bears no such simple relation to the repulsion shape; drawing on the TriMap analysis, we derive the following instead.

Proposition G.3. *The update Eqs. 74-75 provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k, k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l, l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. 74 and 75, $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. 79, we obtain the desired inequality. $\square$

Because t -SNE’s repulsion condition (Eq. 78) mirrors that of TriMap, its repulsion behaves identically: the repulsive forces of t -SNE can produce both attraction and repulsion.

Since t -SNE scales its forces—by $w_{i,j}$ for attraction and by $\frac{w_{i,j}}{Z}$ for repulsion—both are typically weaker than UMAP’s, and the algorithm accordingly adopts large learning rates (for instance $\approx 10^3$ in the Open- t -SNE package (Poličar et al. 2024)). Most t -SNE implementations further rely on an ‘early exaggeration’ step that multiplies the attractive forces by a constant for the first several iterations. This step draws together points that ought to be neighbors but are currently placed far apart, inducing far-sightedness—and, symmetrically, when points sit very close ($\zeta_{i,j} \approx \zeta_{-1}$) yet require separation, it supplies that as well. Early exaggeration is therefore indispensable for a consistent t -SNE embedding, particularly under random initialization; it plays the same role for t -SNE that our modified attraction shape plays for UMAP. Applied throughout optimization rather than only initially, the same device renders t -SNE embeddings visually closer to those of UMAP (Böhm et al. 2022).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis, 2002), one of the earliest algorithms in this class, adopts the same formulation as t -SNE (Eq. 71) but substitutes the Gaussian kernel $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. This choice yields the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

which satisfies Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$), and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which satisfies Proposition G.3 once the SNE quantities are substituted for their t -SNE counterparts. The attraction shape is constant and thus ill-posed as a function of distance, so that contraction depends almost entirely on the learning rate λ and the weights $w_{i,j}$ rather than on the pair’s separation; the consequence is clusters that overlap even for a small number of samples—the crowding problem (Van der Maaten and Hinton, 2008). t -SNE and its successors resolve this by replacing the Gaussian kernel with a heavy-tailed one, namely Eq. 43. Viewed through the shapes, the crowding problem is thus a defect of the attraction shape, and its resolution a matter of reshaping that function.

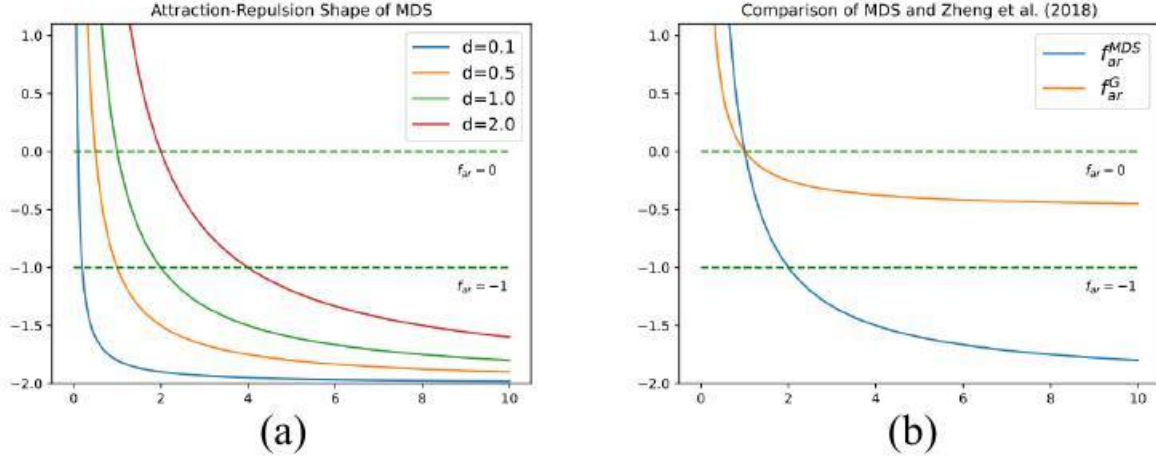


Figure 17: (a) Attraction–repulsion characteristic of MDS as captured by the combined shape f_{ar} . (b) Comparison of the attraction–repulsion characteristics of MDS and the method of (Zheng et al., 2018) at $d = 1.0$, showing how the rescaled shape of the latter falls within the admissible contraction range.

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) typically eschews gradient methods, which often fail to converge to good mappings, in favor of stress majorization; a few works nonetheless pursue gradient formulations (Kruskal, 1964; Zheng et al., 2018), which we treat briefly here. In particular, (Zheng et al., 2018) devise a successful gradient-descent MDS algorithm for graph drawing. We begin from the cost function, absorbing the weight $w_{i,j}$ into the learning rate without loss of generality,

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

This loss contributes a single term per pair, with no explicit split into attractive and repulsive parts; the one term supplies both. Because the loss separates into independent (i, j) components, a single pair may be analyzed in isolation. The derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

Hence the attraction–repulsion shape of MDS is

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

with $\zeta = \|y_i - y_j\|_2$. The shape is attractive whenever $\zeta > d$, over which range it is bounded within $[-2, 0]$ (Fig. 17 a). This range is nonetheless unsuitable for convergence: only values within $[-1, 0]$ contract, the rest expand, and values below -2 flip the pair and expand it further. To circumvent this, (Zheng et al., 2018) adopt an ad hoc formulation inspired by force-directed graph layout,

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which is admissible for any effective learning rate $\lambda \leq 1$, since f_{ar}^G is bounded within $[-0.5, 0]$. For $\zeta < d$, meanwhile, $f_{ar}^{MDS} > 0$ and the interaction is repulsive. This attraction–repulsion interaction performs best when the initialization already lies near the desired output. Continued optimization of a single pair oscillates about the distance $\zeta = d$ at which $f_{ar}^G = 0$, furnishing a quantity analogous to UMAP’s ζ_{-1} ; here, however, no choice of learning rate drives that distance to zero—indeed, holding it at d is the very objective of MDS. MDS clusters are therefore often fuzzier than those of methods such as UMAP and PaCMAP (for illustrations, see (Lambert et al., 2022), (de Bodt et al., 2025), and (Kury et al., 2025)). We note that (Lambert et al., 2022) recast the MDS loss as a quartet stress—a loss over four samples using relative distances, each divided by the sum of the six pairwise distances in the quartet—which enables global regularization at lower computational cost than the majorization approach.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) [Hinton and Roweis \(2002\)](#) underpins the modern dimensionality reduction algorithms. It builds a high-dimensional graph of the dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$ through the system

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is set to match a user-specified *perplexity* P , defined by

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP, by contrast, constructs its high-dimensional graph from the k -nearest-neighbor (k -NN) graph through the system

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k nearest neighbors of x_i and σ_i is a scaling parameter chosen so that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized via a t-conorm,

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP relies solely on the k -NN graph for its affinities—all set to 1—together with a self-tuning distance measure [Zelnik-Manor and Perona \(2004\)](#),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance from x_i to its fourth through sixth Euclidean nearest neighbors. This σ_i serves the same role as its counterparts in Eqs. [\(87\)](#) and [\(90\)](#), even though the three are defined differently.

These graph-construction choices differ substantially across methods; our analysis nonetheless indicates that, once the graph is fixed, the optimization—and specifically the attraction and repulsion shapes—is the primary determinant of the low-dimensional embedding.

I Detailed Results for Varying λ_a and λ_b

This section reports the detailed results underlying the experiments of Fig. 11 h,i), obtained by substituting the attraction and repulsion shapes of the other methods for UMAP’s within a single common implementation. Figure 18 reproduces the main-text results. Figures 19-22 present the individual embeddings for each choice of λ_a and λ_r per method, alongside the initialization, a reference embedding obtained under annealed λ , and the two degenerate cases in which λ_a or λ_r is set to 0. For PaCMAP, whose formulation includes mid-near pairs, we additionally show a reference embedding with those pairs included. Figures 23-27 and Figures 28-32 give the corresponding results on the FMNIST and single-cell transcriptomes datasets. Across all three datasets the pattern is the same: repulsion alone and attraction alone each fail to form clusters, and only UMAP demands a small λ_a .

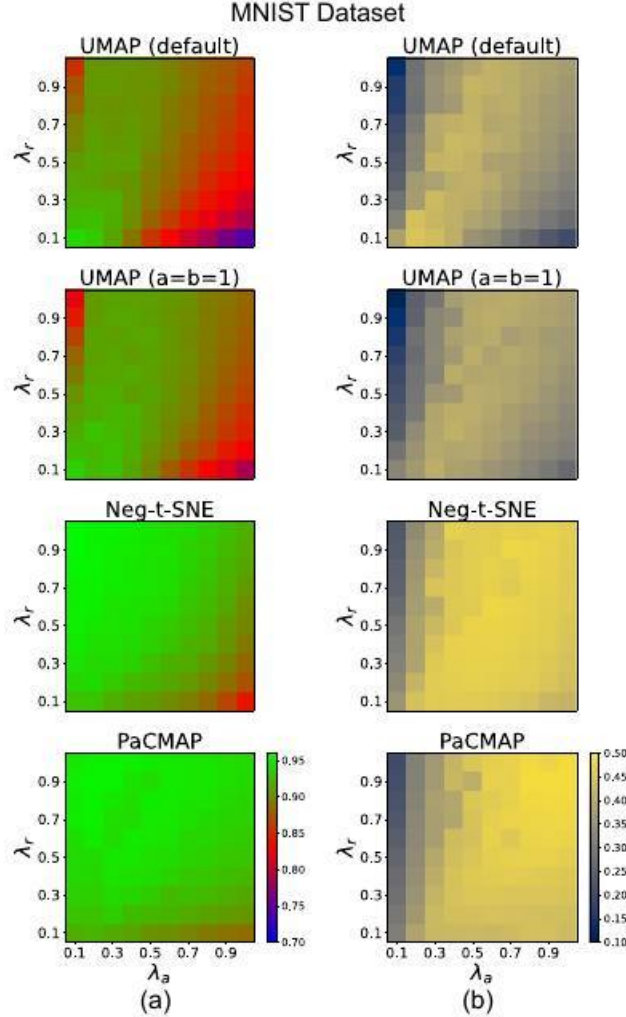


Figure 18: (a) Trustworthiness and (b) silhouette score across methods on the MNIST dataset, as the attraction learning rate λ_a and repulsion learning rate λ_r are varied independently. Of the methods compared, only UMAP requires a small λ_a to embed well.

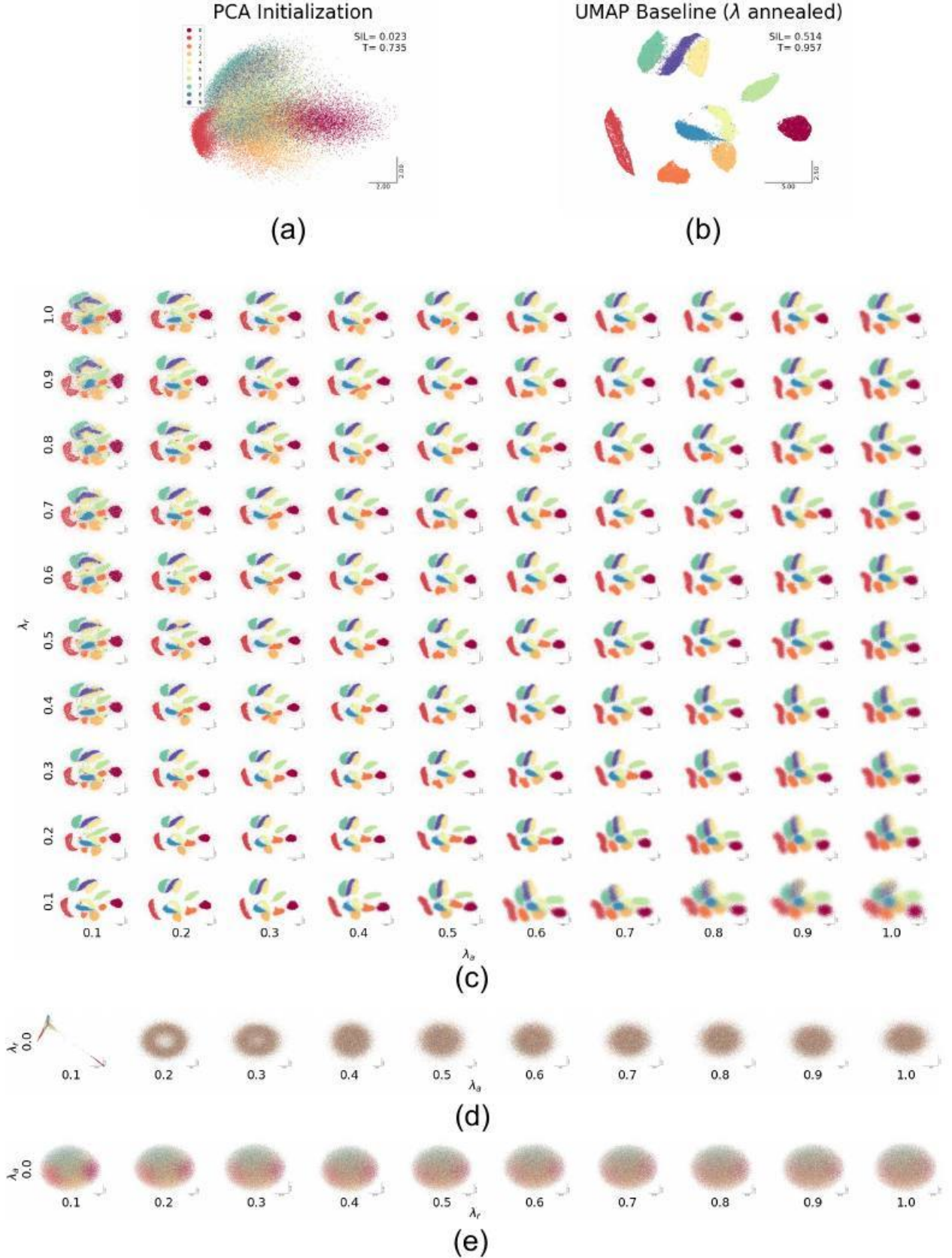


Figure 19: Varying λ_a and λ_r on the MNIST dataset with UMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline obtained with λ annealed from 1. (c) Embeddings across values of λ_a and λ_r at constant learning rate, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters. Panels (d,e) together confirm that both forces are jointly necessary for cluster formation.

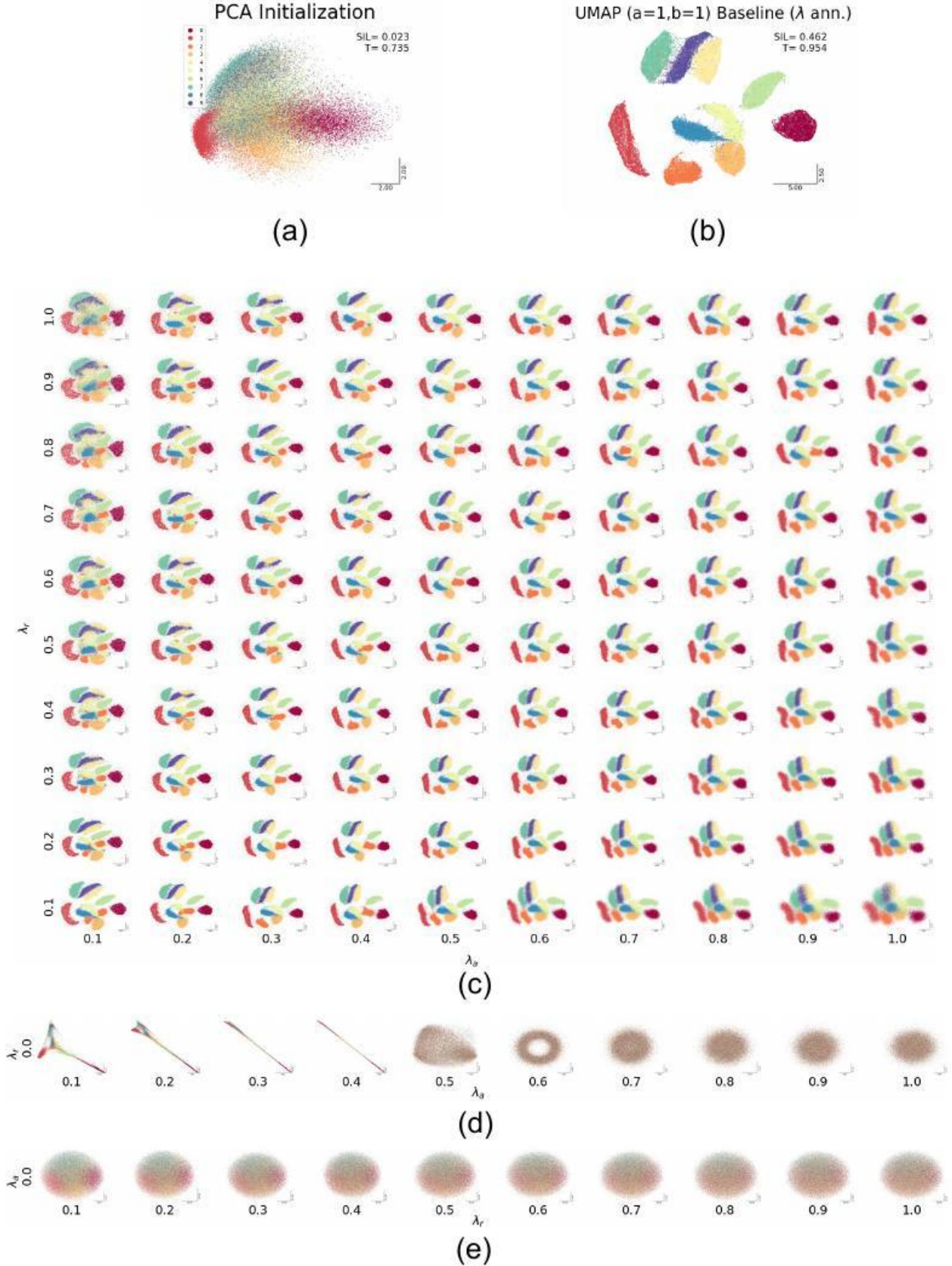


Figure 20: Varying λ_a and λ_r on the MNIST dataset with UMAP's attraction and repulsion shapes in the unit-parameter case ($a = 1$, $b = 1$). (a) Initialization of the embeddings. (b) Baseline obtained with λ annealed from 1. (c) Embeddings across values of λ_a and λ_r at constant learning rate, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

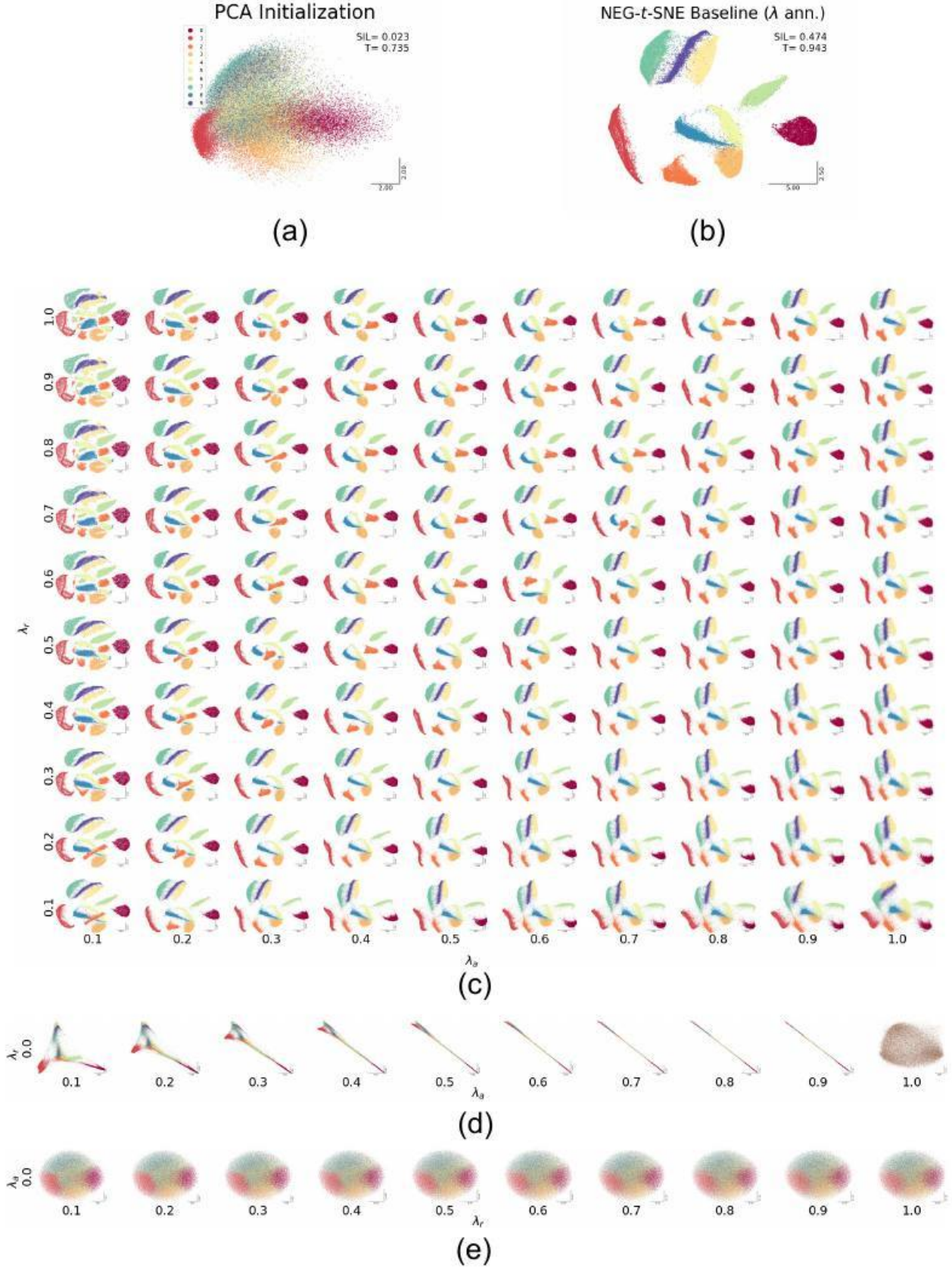


Figure 21: Varying λ_a and λ_r on the MNIST dataset with NEG-t-SNE's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline obtained with λ annealed from 1. (c) Embeddings across values of λ_a and λ_r at constant learning rate, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

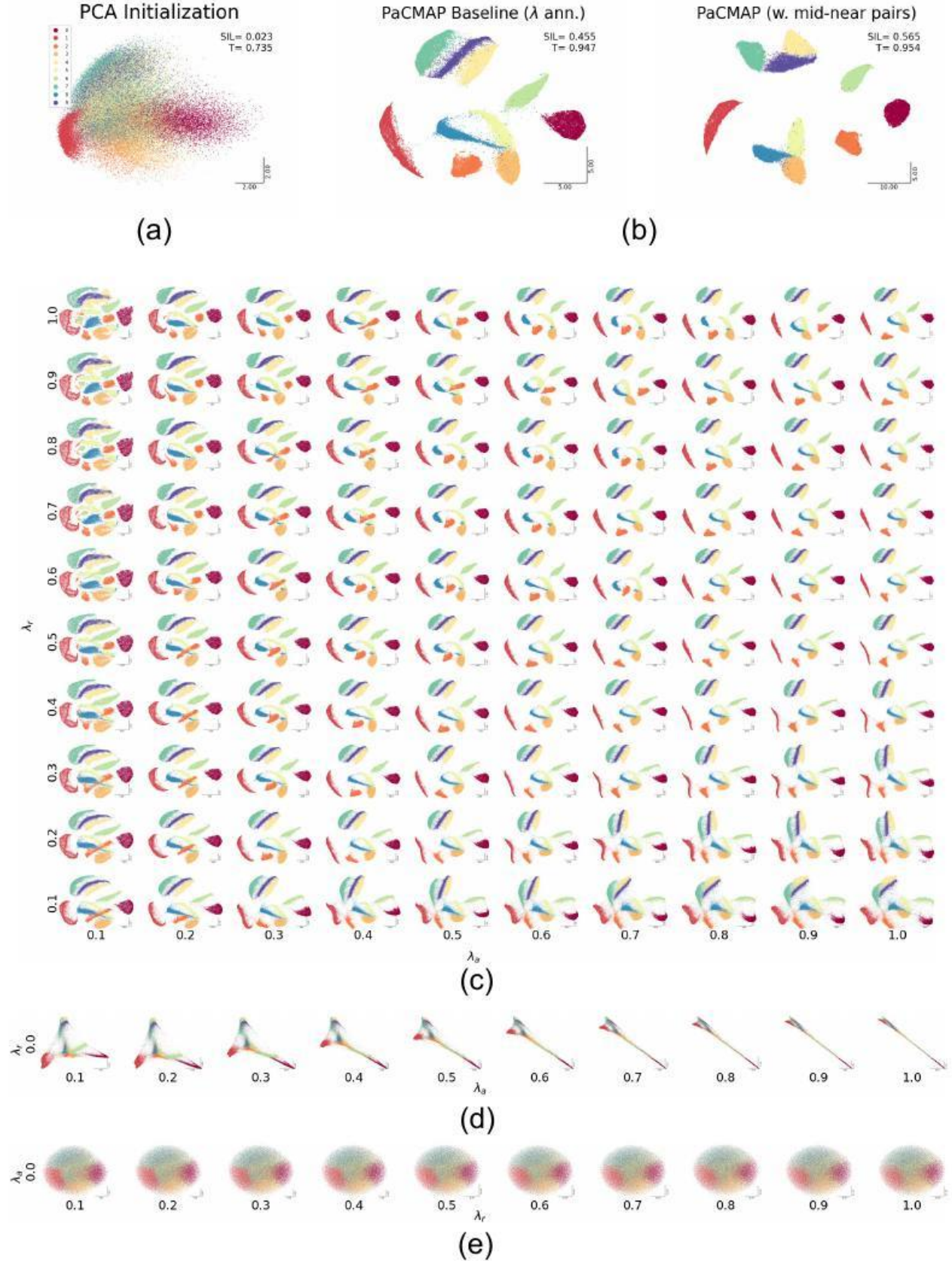


Figure 22: Varying λ_a and λ_r on the MNIST dataset with PaCMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Left: baseline with λ annealed from 1 and mid-near pairs excluded, isolating the attraction-repulsion interaction; right: the same with mid-near pairs included. (c) Embeddings across values of λ_a and λ_r at constant learning rate, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

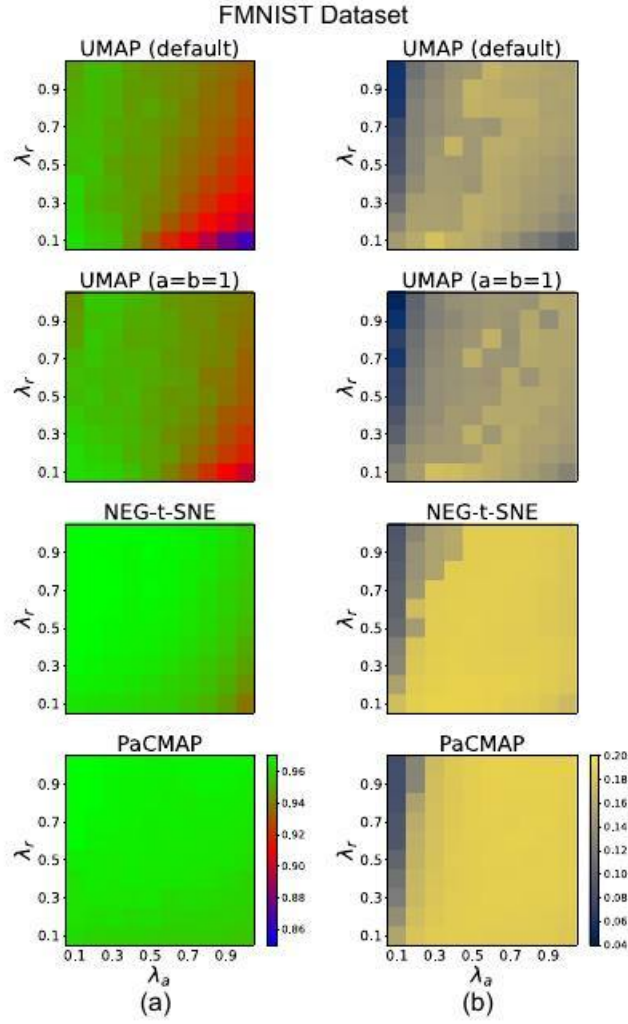


Figure 23: (a) Trustworthiness and (b) silhouette score across methods on the FMNIST dataset, as the attraction learning rate λ_a and repulsion learning rate λ_r are varied independently, replicating the MNIST comparison of Fig. 18

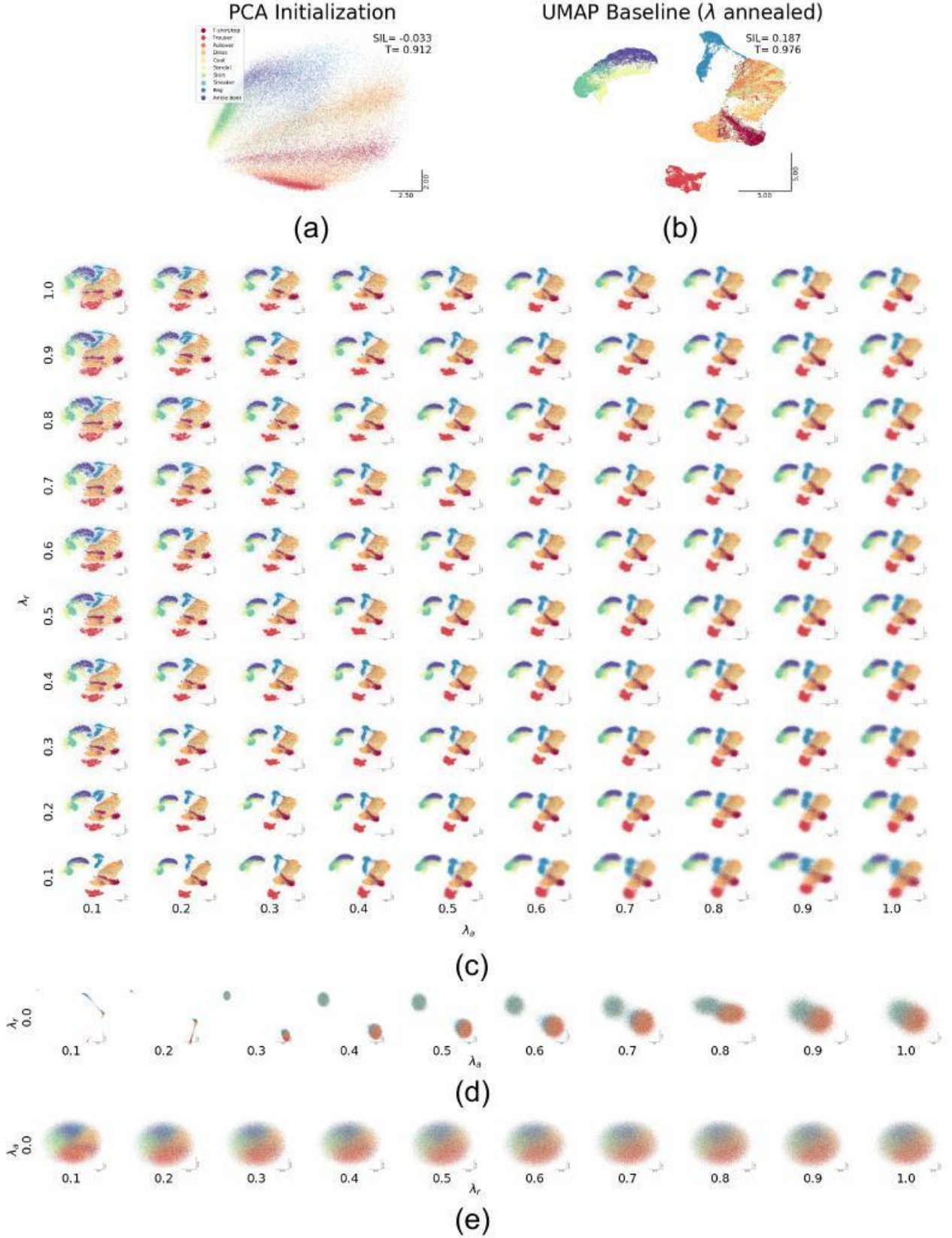


Figure 24: Varying λ_a and λ_r on the FMNIST dataset with UMAP’s attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline obtained with λ annealed from 1. (c) Embeddings across values of λ_a and λ_r at constant learning rate, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

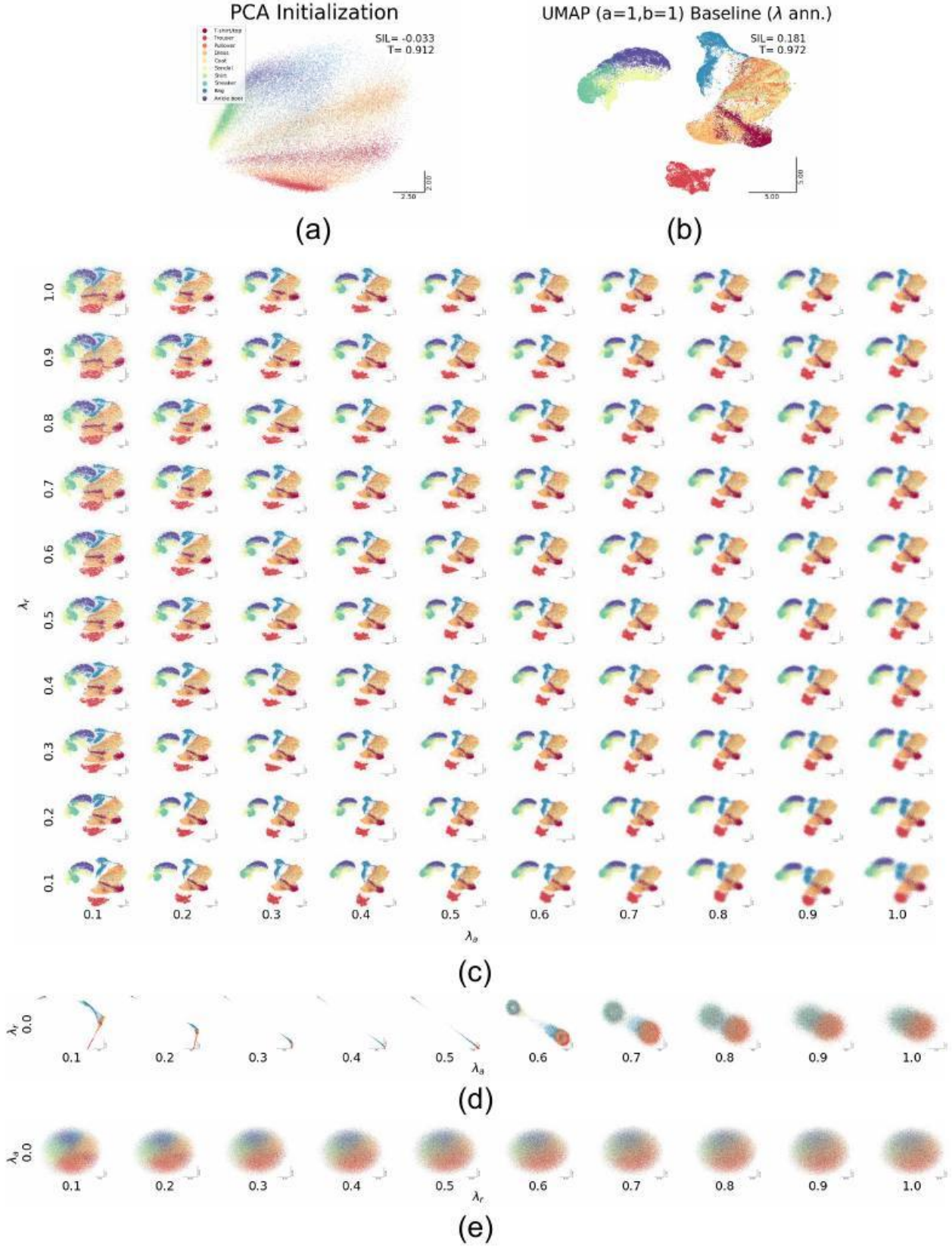


Figure 25: Varying λ_a and λ_r on the FMNIST dataset with UMAP’s attraction and repulsion shapes in the unit-parameter case ($a = 1, b = 1$). (a) Initialization of the embeddings. (b) Baseline obtained with λ annealed from 1. (c) Embeddings across values of λ_a and λ_r at constant learning rate, without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

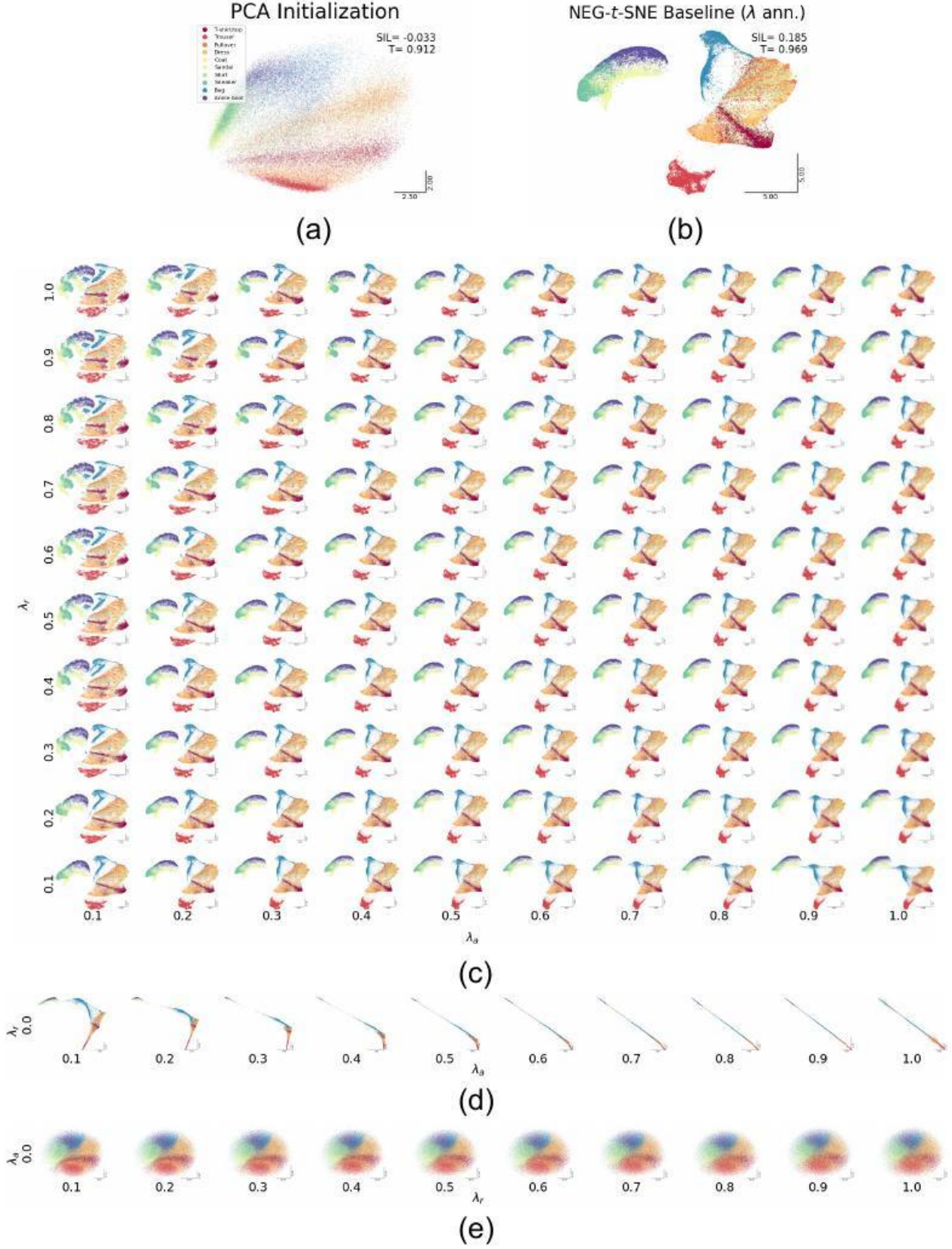


Figure 26: Varying λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline obtained with λ annealed from 1. (c) Embeddings across values of λ_a and λ_r , without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

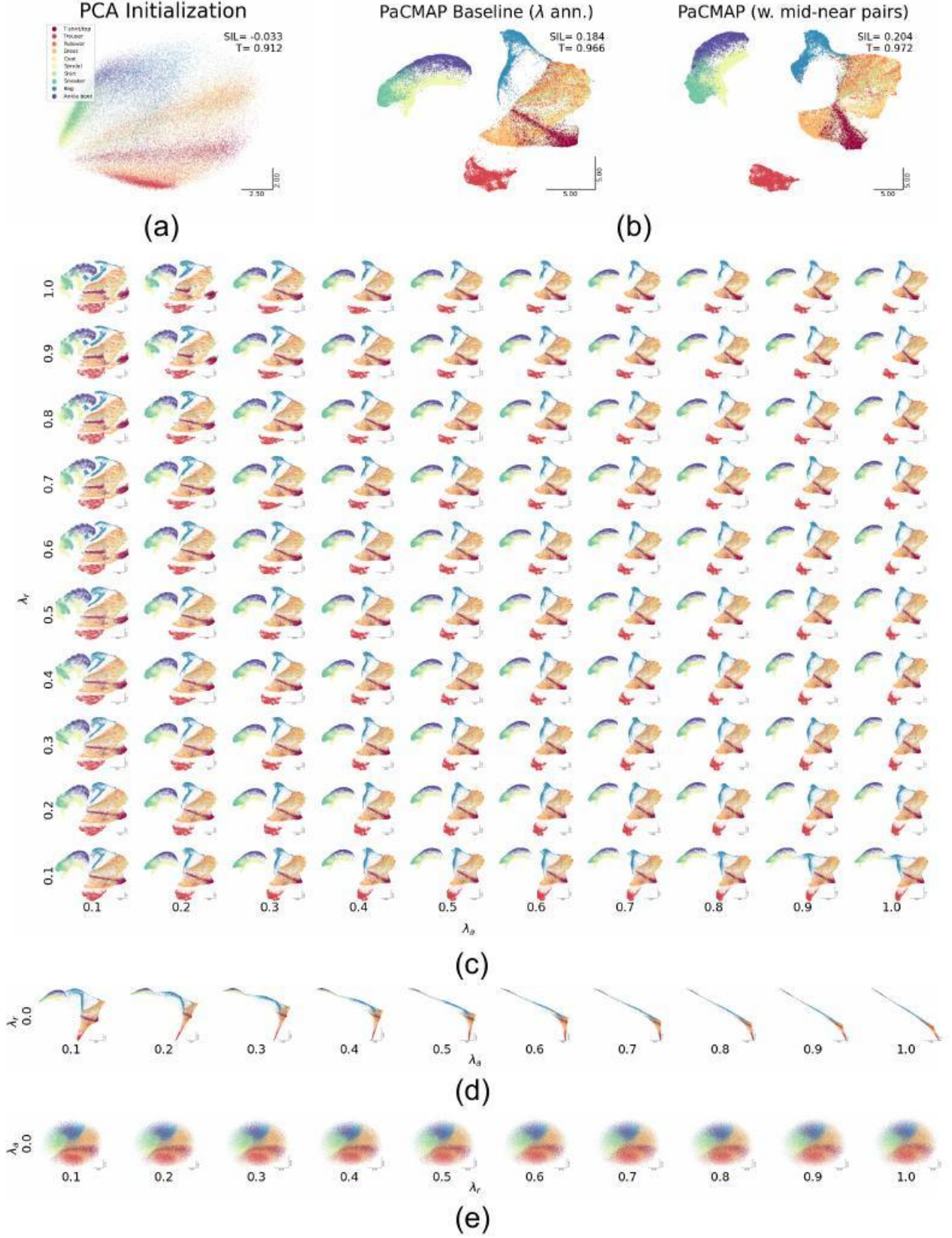


Figure 27: Varying λ_a and λ_r for the FMNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Left: baseline with λ annealed from 1, mid-near pairs excluded so as to isolate the attraction-repulsion interaction; right: the same with mid-near pairs included. (c) Embeddings across values of λ_a and λ_r , without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

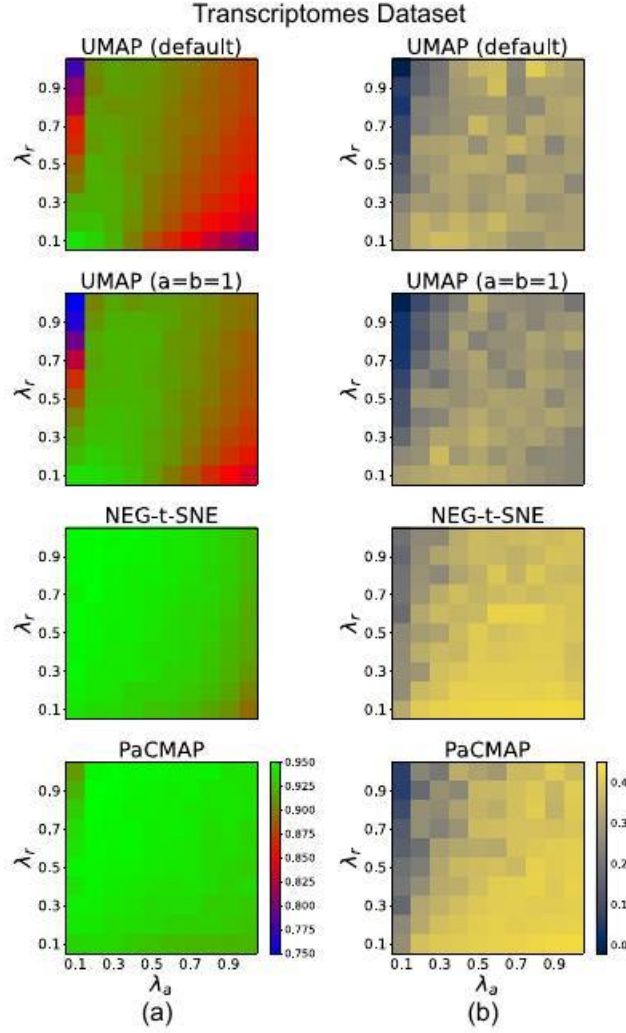


Figure 28: (a) Trustworthiness and (b) silhouette score across methods on the single-cell transcriptomes dataset, as the attraction learning rate λ_a and repulsion learning rate λ_r are varied independently, extending the comparison of Fig. 18 to a biological dataset.

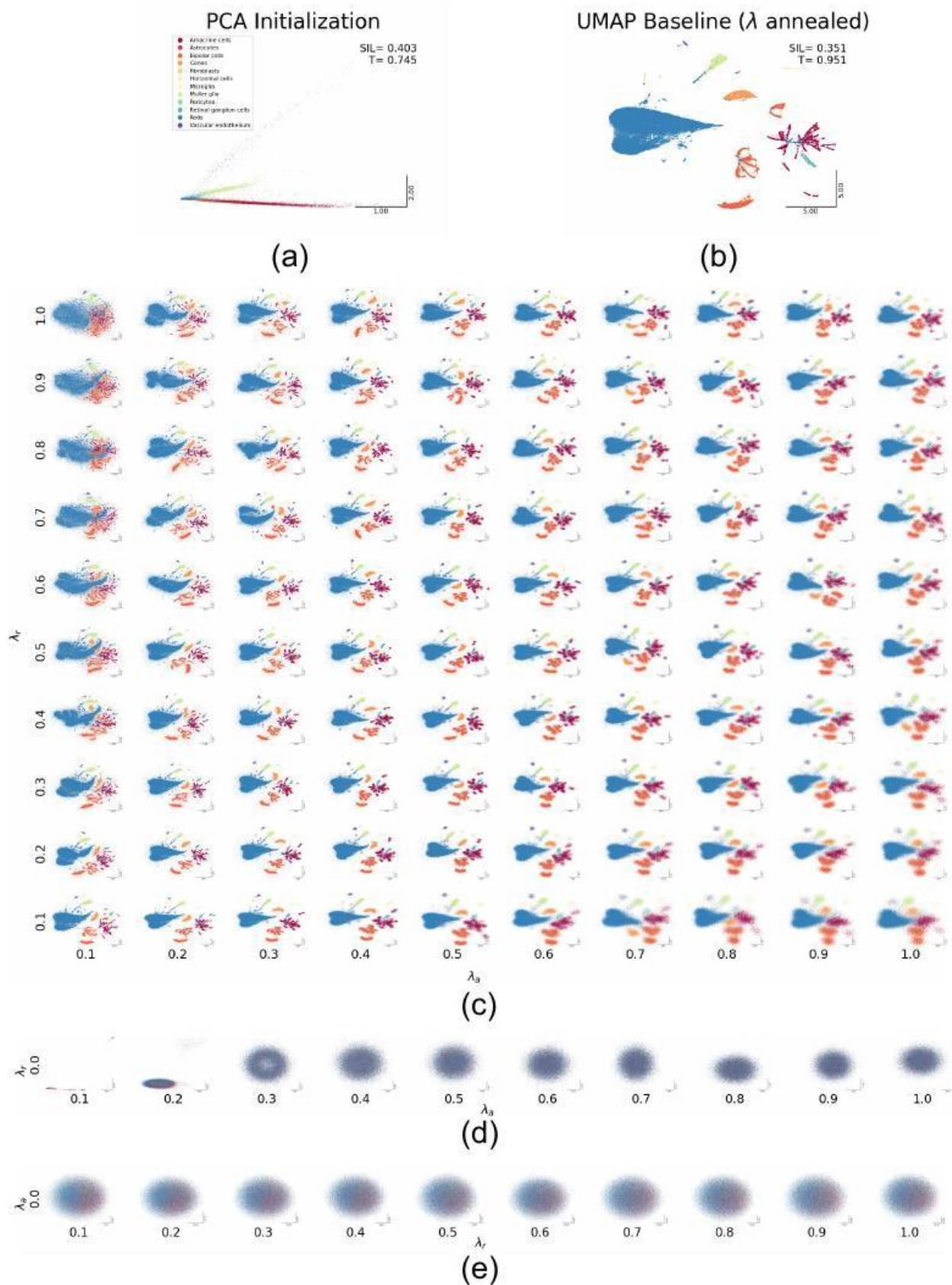


Figure 29: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Baseline obtained with λ annealed from 1. (c) Embeddings across values of λ_a and λ_r , without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

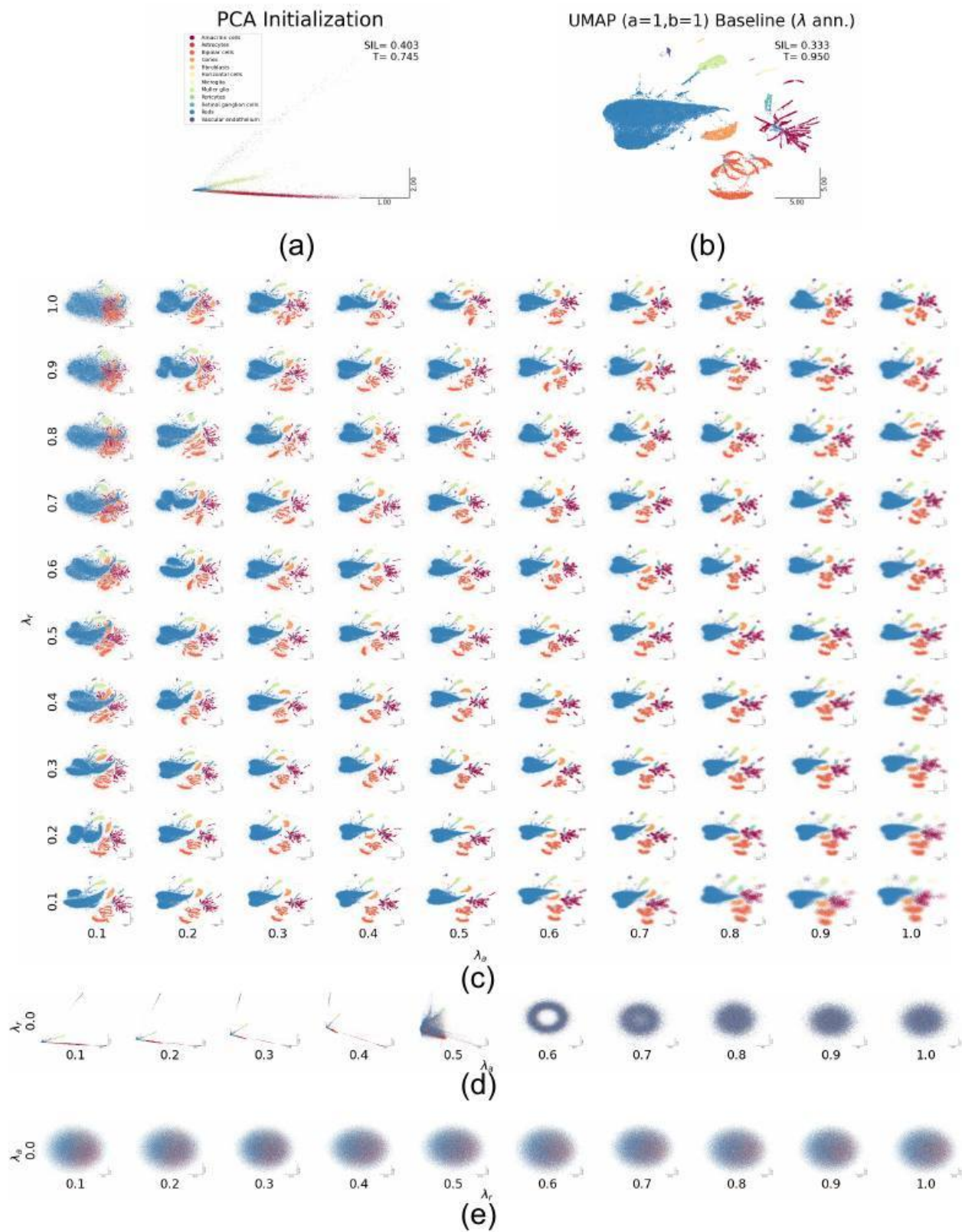


Figure 30: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), repulsive force alone cannot produce any clusters.

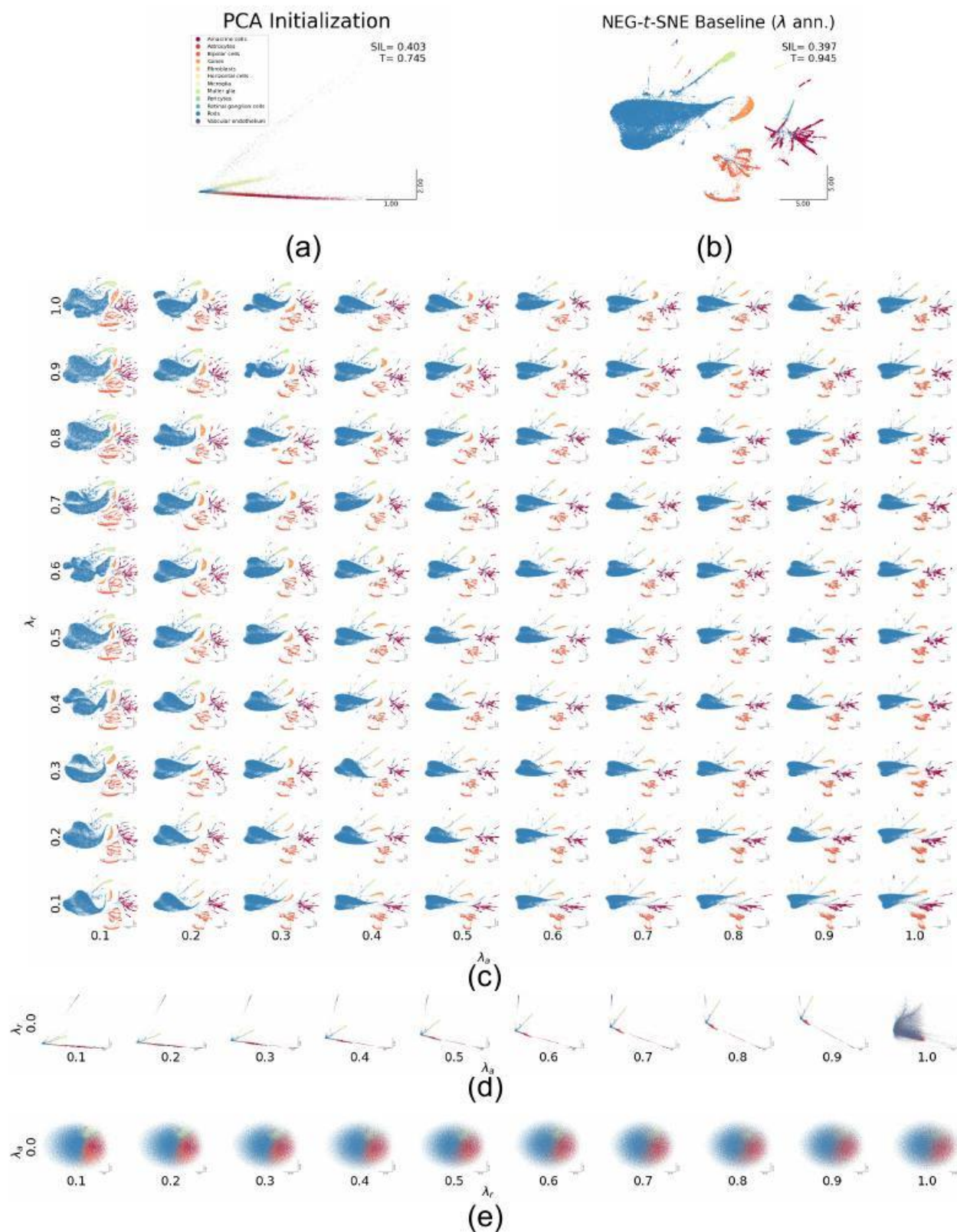


Figure 31: Varying λ_a and λ_r for the single-cell transcriptomes dataset using NEG-*t*-SNE’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce distinct clusters. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

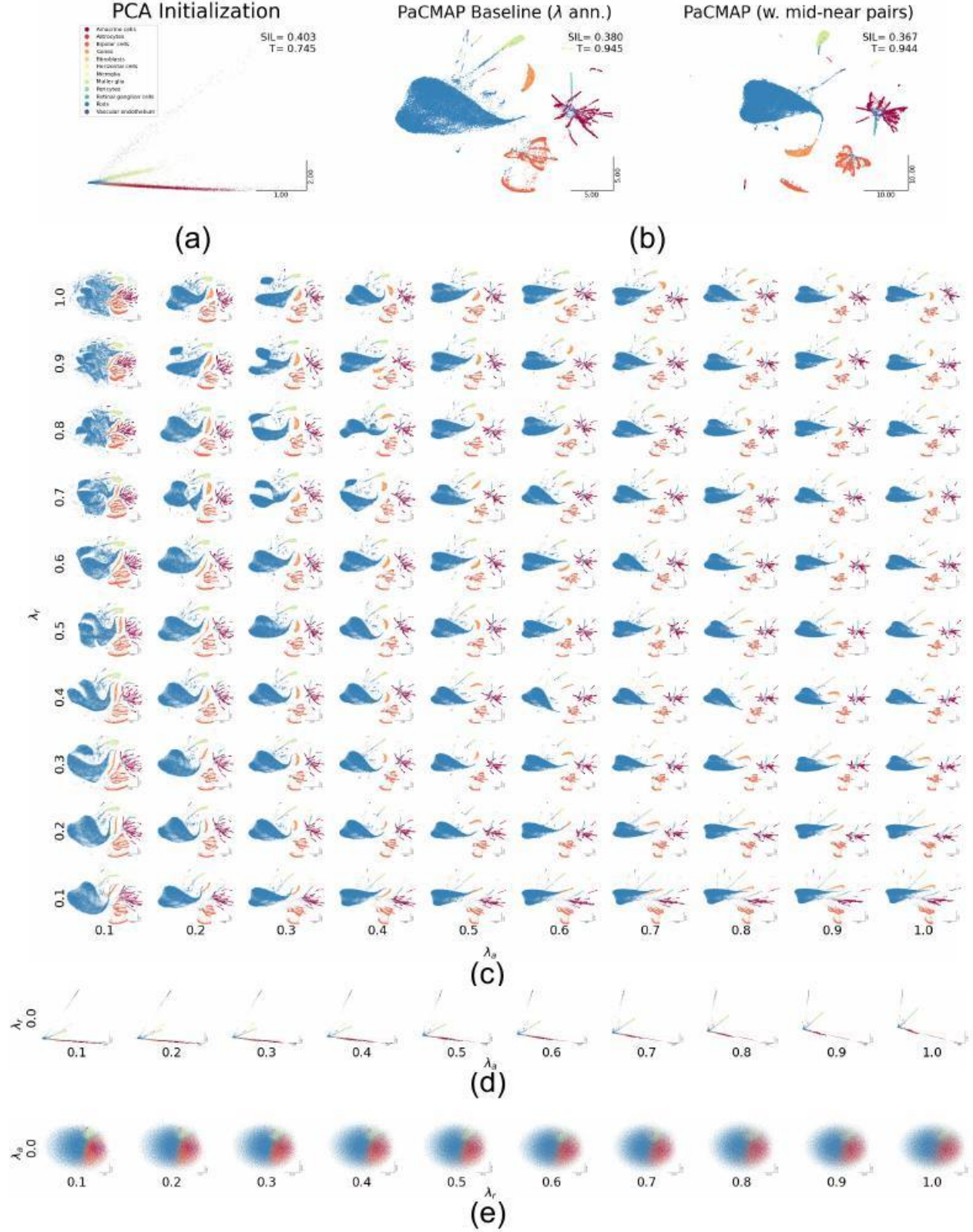


Figure 32: Varying λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization of the embeddings. (b) Left: baseline with λ annealed from 1, mid-near pairs excluded so as to isolate the attraction–repulsion interaction; right: the same with mid-near pairs included. (c) Embeddings across values of λ_a and λ_r , without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone produces no clusters. (e) Symmetrically, with $\lambda_a = 0$ (no attraction), repulsion alone produces no clusters.

J Mix and Match Approach

Because our analysis treats the attraction and repulsion shapes as independent, it follows directly that shapes derived from different loss functions and principles can be freely combined. Figure 33 presents several such combinations. In each, we confined the attraction shapes to $[-1, 0]$ (to comply with Proposition 4.1), rendering them insensitive to the learning rate, and used finite repulsion shapes to suppress the large repulsive forces that would otherwise arise at small distances. Holding the learning rate constant at $\lambda = 1$ and applying the identical UMAP optimization throughout ensures that any observed differences stem from the shapes alone, not from learning-rate scheduling or the optimization scheme. Two clear patterns emerge: UMAP’s attraction shape (via $f_a = 0.5f_a^U$) yields better-separated clusters, reflected in a higher silhouette score, whereas PaCMAP’s attraction shape yields better local structure, reflected in higher trustworthiness—the latter owing to PaCMAP’s attraction shape lying within $[-0.5, 0]$, which forbids flips during the updates. Confining the attraction shapes of UMAP and NEG- t -SNE to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively) trades clustering performance for improved local structure (Fig. 34).

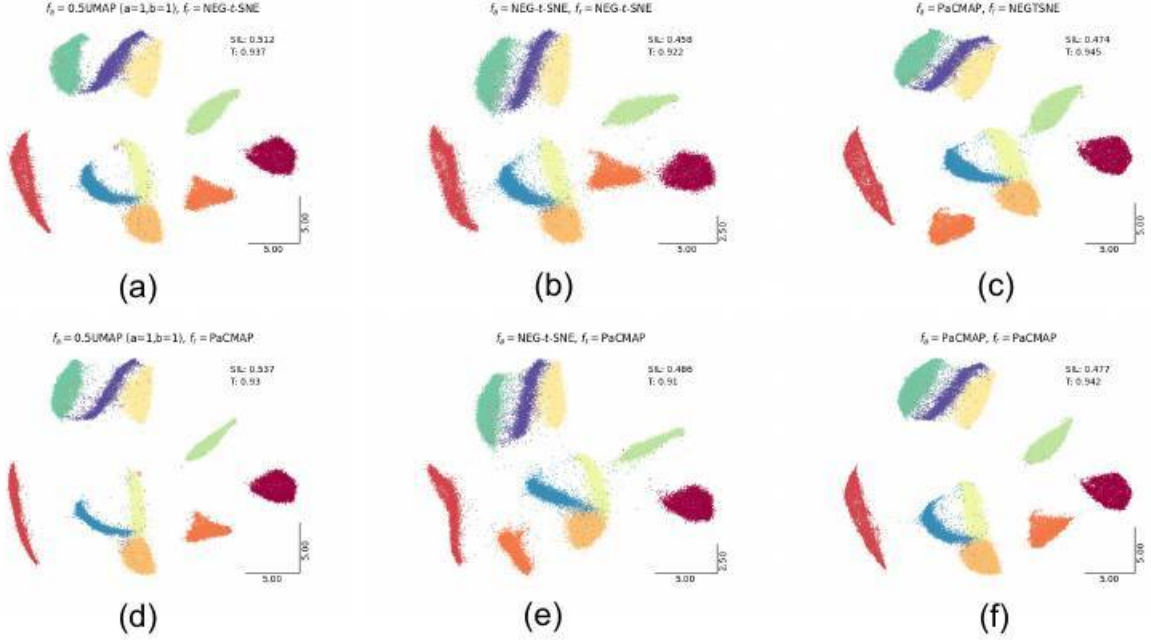


Figure 33: Embeddings of MNIST obtained by mixing and matching attraction and repulsion shapes at a constant learning rate ($\lambda = 1$), using attraction shapes that satisfy Proposition 4.1 and finite repulsion shapes. Top row: the attraction shapes of UMAP, NEG- t -SNE, and PaCMAP, each paired with NEG- t -SNE’s repulsion shape. Bottom row: the same attraction shapes, each paired with PaCMAP’s repulsion shape.

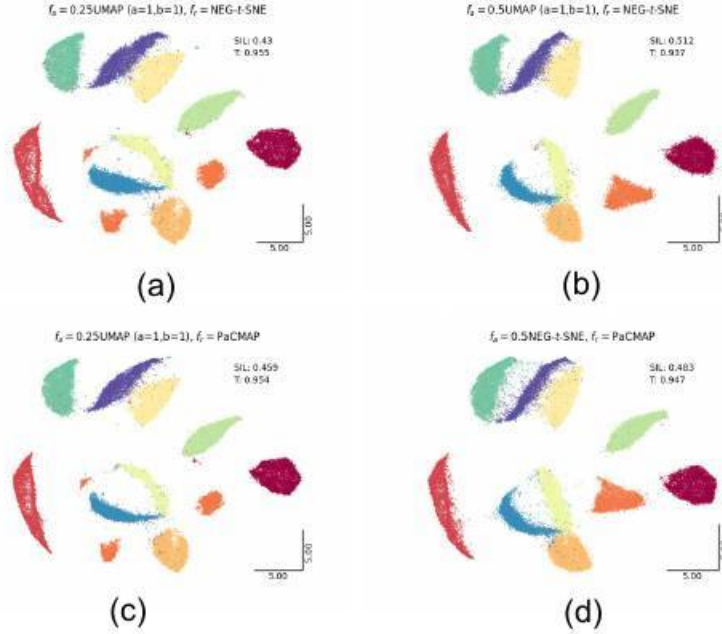


Figure 34: Embedding of MNIST by mixing and matching attraction and repulsion shapes with a constant learning rate ($\lambda = 1$). We kept the attraction shape confined to $[-0.5, 0]$ to prevent any flips during the attractive update. Top row: UMAP and NEG- t -SNE’s attraction shapes with NEG- t -SNE’s repulsion shape. Bottom row: UMAP and NEG- t -SNE’s attraction shapes with PaCMAP’s repulsion shape.

K Implementation Details

For analysis, we implemented our own UMAP algorithm. We used `numba` (Lam et al., 2015) to compute an exact nearest neighbor graph (instead of an approximate one) with $k = 15$ and `scikit-learn`'s (Pedregosa et al., 2011) implementation of the PCA algorithm for PCA initialization. We used this to produce and quantify the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9 and 10. We also used the same implementation when we changed the attraction and the repulsion shapes to those of the alternative methods (Figs. 18, 34, unless otherwise stated). The trustworthiness and silhouette scores were computed using the corresponding function from the `scikit-learn` package.

The mappings shown in Figs. 2, 3, 9, 10 and 15 are rotated to a reference embedding ((a) for each respective Figures). To achieve this, we performed Procrustes alignment of the embeddings by normalizing them (zero mean and unit norm) and then using SciPy's (Virtanen et al., 2020) `orthogonal_procrustes` method to extract rotation and scaling parameters.

To compare with Neg- t -SNE in Fig. 11, we used the original implementation of the contrastive embedding framework for both the UMAP and the Neg- t -SNE algorithms; the implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22(b) (right), 27(b) (right), and 32(b) (right) were obtained using the official PaCMAP package; the implementation details will be released upon acceptance.

The codes for displaying the embeddings of the PubMed dataset in Fig. 7 will be released upon acceptance.